Contents

1 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of intrest for mathematicians and physisists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 1.0.1 (Topological Manifold)**.** A second countable Hausdorffspace M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around p on M** and φ^{-1} a **local coordinate system around p on M**.

Definition 1.0.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension M.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i,j) \in I^2$ the **transition function**:
   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 1.0.3. Every transition functions φ_{ij} where $(i,j) \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$

Definition 1.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuos function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any one (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f \in U \to \mathbb{R} \text{ continouos} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 1.0.5. Let (M,c) be a smooth manifold of dimension m and $U \subseteq M$ be a open subset. With pointwise defined operations, the set $(\mathcal{E}(U),+,\cdot,\circ)$ becomes an $\mathbb{R}$–algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \mathrm{res}_V^U : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V \,. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. ☐

Definition 1.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_w = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\bigsqcup_{U \mathrm{open}, p \in U} \mathcal{E}(U) \right) \Big/ \sim_p \,.$$

For the set of germs and call it the **stalk in** p. Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 1.0.7. For a smooth manifold (M,c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$–algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \mathrm{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\mathrm{eval}_p)$. Hence, the pair $(M,\mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\mathrm{eval}_p)$. ☐

Definition 1.0.8. Let (M,c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f,g \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})\,.$$

Corollary 1.0.9. Let (M,c) be a smooth manifold and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ ($\varphi \in \mathfrak{A} \in c$). Then

$$\xi = \left.\frac{\partial}{\partial x_j}\right|_p : \mathcal{E}_p(M) \to \mathbb{R}$$
$$f_p \mapsto \xi(f_p) = \left.\frac{\partial}{\partial x_j}\right|_{x_0} (f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\left.\frac{\partial}{\partial x_1}\right|_p, \dots, \left.\frac{\partial}{\partial x_m}\right|_p \right)$$

defines a basis of TM_p. Hence, the dimension of TM_p is m.

Definition 1.0.10. For a smooth manifold (M,c) the sum $\bigsqcup_p TM_p$ comes with a natural projection

$$\pi : TM \to M$$
$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \to V$

$$\frac{\partial}{\partial x_j} : U \to \pi^{-1}(U)$$
$$p \mapsto \left.\frac{\partial}{\partial x_j}\right|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m .$$

We can induce a topology on TM such that all those trivializations are continuos. This then gives an atlas for TM such that we have a $2m$-dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \to R^m \times R^m; x \mapsto (\pi(x), q_{\varphi\circ\pi(x)}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\left.\frac{\partial}{\partial x_1}\right|_p, \dots, \left.\frac{\partial}{\partial x_m}\right|_p)$ that depends on the chart φ_i. In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m. We call TM the **tangent bundle**.

Definition 1.0.11. For a m dimensional vector space V with basis $\mathfrak{b}$ we define the two maps:

$$B_{\mathfrak{b}} : \mathbb{R}^m \to V : \quad e_i \mapsto \mathfrak{b}_i$$
$$q_{\mathfrak{b}} : B_{\mathfrak{b}}^{-1} : V \to \mathbb{R}^m .$$

Definition 1.0.12 (The Derivative)**.** Let (M,c) and (M',c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth function between the tangent bundles as follows:

$$Df : TM \to TM' \ , Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_pM$, $g_p \in E_p(M')$.

Corollary 1.0.13. Let $f : M \to \mathbb{R}$ be smooth and $\varphi : U \to V$ be a chart. Then we can interpret $\mathrm{d}f$ as a one-form. To be precise assume q to be the coordinate function $T\mathbb{R} \to \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ \mathrm{d}f(v) = v \cdot f$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notations so vector fields can take functions as an input.

Proof. We proof this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M,\mathbb{R})$ vector space. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f(\frac{\partial}{\partial x_i}\Big|_p)(g_p) = \frac{\partial}{\partial x_i}\Big|_p (g \circ f)_p = \frac{\partial}{\partial x_i}\Big|_{x_0} (g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\Big|_p (g_p) \cdot \frac{\partial}{\partial x_i}\Big|_p (f_p)$$

Hence, $\mathrm{d}f(v) = v(f)\frac{\partial}{\partial x}$ letting us conclude the statement. □

Definition 1.0.14. Let V be a linear manifold (i.e. a finite dimensional vector space). Then for any $v \in V$ we have the canonical maps

$$\begin{aligned} Q_v : V &\to T_vV \\ x &\mapsto \left(f_v \mapsto \frac{\partial}{\partial t}\Big|_{t=0} f(v+tx) \right) \end{aligned}$$

Let $L : V \to V$ be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$\mathrm{d}L|_v \circ Q_v = Q_{L(v)} \circ L\,.$$

This is proven by a short calculation:

$$\Big(\mathrm{d}L|_v \left(Q_v(x)\right)\Big)(f_{L(v)}) = Q_v(x)\,(f \circ L) = \frac{\partial}{\partial t}\Big|_{t=0} f \circ L(v+tx) = \underbrace{\frac{\partial}{\partial t}\Big|_{t=0} f\big(L(v) + tL(x)\big)}_{=Q_{L(v)} \circ L(x)}$$

Now let $v, v' \in V$ be two points in V. Then we have the translation

$$tr_{v,v'} : V \to V;\; x \mapsto (x + (v' - v))$$

With this we can now find a connection between Q_v and $Q_{v'}$ as follows:

$$\left(\mathrm{d}tr_{v,v'}|_v\left(Q_v(x)\right)\right)(f_{v'}) = \frac{\partial}{\partial t}\bigg|_{t=0}(f \circ tr_{v,v'}(v + tx)) = \frac{\partial}{\partial t}\bigg|_{t=0}\left(f(v' + tx)\right) = Q_{v'}(x)(f_{v'})\,.$$

Hence:

$$Q_{v'} = \mathrm{d}tr_{v,v'} \circ Q_v\,.$$

The map Q then denotes the map on the whole tangent bundle:

$$Q : TV \to V; \xi_p \mapsto Q_p(\xi_p)$$

Definition 1.0.15 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual tangend bundle with itself is a **Riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_pM$ $g_p(v_p, w_p) = 0$ $\forall w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ $\forall v_p, w_p \in T_pM$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0 \forall v_p$ and vanishes only for $v_p = 0$.

We cal a tuple (M, g) a **Riemannian manifold**.

Definition 1.0.16 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a Riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vecor field V:

$$Q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V)\,.$$

Definition 1.0.17. Let (M, g) be a Riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be :

$$g_{ij} = g(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j})\,.$$

Then if two vector fields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v, w) = \sum_{ij} g_{ij} v_i w_j \;: U \to \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via Cramers rule) to get smooth functions

$$\begin{aligned} g^{ij} &: U \to \mathbb{R} \\ &p \mapsto g^{ij}(p) = \left((g_{kl}(p))^{-1}_{kl}\right)_{ij} \end{aligned}$$

Remark 1.0.18. Due to the rare appearance of big summations with vectors and co-vectors in this work, we omit the Einstein summation convention and give all coordinate entries lower indices.

Lemma 1.0.19. *Let $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\operatorname{grad}(f) = \sum_{i,j} g^{ij}\left(\frac{\partial}{\partial x_i}(f)\right)\frac{\partial}{\partial x_j} =: \sum_{i,j} g^{ij}\frac{\partial f}{\partial x_i}\frac{\partial}{\partial x_j}\,.$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum_i v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$. Then $g(v,w) = \sum_{i,j} g_{ij} v^i w^i$. Suppose that $\operatorname{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \to \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ \mathrm{d}f(\frac{\partial}{\partial x_j}) = q \circ \frac{\partial f}{\partial x_j} = g(\operatorname{grad}(f), \frac{\partial}{\partial x_j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$\begin{aligned}(G^1, \cdots G^m)(g_{ij}) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right)\\ \Leftrightarrow (G^1, \cdots G^m) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right)(g^{ij})\,.\end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij} \frac{\partial}{\partial x_i}(f)$.

□

Definition 1.0.20 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$

3. For all p we require $\varphi(0,p) = p$.

We will write $\varphi_t(p) := \varphi(t,p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 1.0.21. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff},\ t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t.

Definition 1.0.22 (Vector Field of a Dynamical System)**.** Let φ be a dynamical system on a smooth manifold M. Then we call the mapping

$$p \mapsto X(p) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \varphi_t(p) \in T_pM$$

the **associated vecor field**. This is in fact a smooth vector field.

Definition 1.0.23 (Dynamical System of a Vector Field)**.** Let $X \in \Gamma(M)$ be a global vector field on the smooth manifold M. Then we call a dynamical system φ on M the **associated dynamical system** to X, if for all $t \in \mathbb{R}$ and $x \in M$

$$\mathrm{d}_t\varphi_t(x)|_{t=t_0} = X(\varphi_{t_0}(x)) \, .$$

d_t denotes the differential in t, meaning for fixed x we differentiate

$$\varphi(x) : \mathbb{R} \to M, \quad t \mapsto \varphi_t(x)$$

and evaluate that in the constant trivial vector field $E : \mathbb{R} \to T\mathbb{R}$ with $E(t) = 1_t$.

Those two concepts are inverse to one another or in other words:

Theorem 1.0.24. *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

Remark 1.0.25. Let $X \in \Gamma(TM)$ be a vector field on the smooth manifold M, and φ be the associated dynamical system. For any fixed $t_0 \in \mathbb{R}$ and any $x_0 \in M$ we have the identity

$$\mathrm{d}\varphi_{t_0}|_{x_0} \circ (X(x_0)) = X(\varphi_{t_0}(x_0)) \, .$$

Proof. This follows directly from the second property of flows as follows: We first make use of the identity $\varphi_{t_0}(x_0) = \varphi_{t_0} \circ \varphi_0(x_0)$ and $X(x_0) = \mathrm{d}_t\varphi_t(x_0)|_{t=0}$ to calculate using the chain rule:

$$\mathrm{d}_t\varphi_{t_0+t}(x_0)|_{t=0} = \mathrm{d}_t\varphi_{t_0} \circ \varphi_t(x_0)|_{t=0} = \mathrm{d}_x\varphi_{t_0}|_{x=\varphi_0(x_0)} \circ \mathrm{d}\varphi_t(x_0)|_{t=0} = \mathrm{d}_x\varphi_{t_0}|_{x=x_0} \circ X(x_0) \, ,$$

and alternatively

$$\mathrm{d}_t\varphi_{t_0+t}(x_0)|_{t=0} = \mathrm{d}_t\varphi_t(\varphi_{t_0}(x_0))|_{t=0} = X(\varphi_{t_0}(x_0)) \, .$$

□

2 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse theory, we will always deal with a smooth function $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 2.0.1 (Critical Points)**.** We call a point $p \in M$ **critical**, if $\mathrm{d}f|_p$ vanishes meaning that

$$\mathrm{d}f|_p : T_pM \to T_{f(p)}\mathbb{R}$$

is trivial. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 2.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear function

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangent vectors. Now choose two vector fields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))\,(p)$$

Compare 1.0.13 for the notation.In corollary 2.0.5 we will give a local description telling us that his definition is well defined, meaning independent of the Extensions Ξ and Z.

Definition 2.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a symmetric bilinear and non-degenerate Map. For such a map there exists the following theorem:

Theorem 2.0.4 (Sylvesters Law of Inertia)**.** *Let* $A \in \mathrm{Mat}(n, \mathbb{R})$ *be symmetric and let* $T, T' \in \mathrm{GL}(n, \mathbb{R})$ *and* $k, k', l, l' \in \mathbb{N}$ *such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ *and* $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^t \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^t \circ A \circ T') = k' + l'\,. \tag{1}$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^t Ax > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing „$\leq$“: Denote the first k colons of T with $x_1, ..., x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda i x_i$ we have that by bilinearity

$$x^t Ax = \sum_{i=1}^{k} \lambda_i x_i^t Ax = \sum_{i,j=1}^{k} \lambda_i \lambda_j x_i^t Ax_j = \sum_{i=1}^{k} (\lambda_i)^2 \geq 0\,. \tag{2}$$

This concludes the first inequality.

Now let U be any $k-$dimensional subspace such that for all non zero $x \in U$ $x^t Ax > 0$. By a calculation analog to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots x_n)$ the number $x^t Ax$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k\,. \tag{3}$$

This is the last inequality proving the statement. □

Corollary 2.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corrsponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)_{i,j}$$

Proof. This is just an easy calculation, since $\operatorname{Hess}_p(f)\left(\frac{\partial}{\partial x_i}\Big|_p, \frac{\partial}{\partial x_j}\Big|_p\right)$ is given by

$$\left(\frac{\partial}{\partial x_i}\Big|_p \left(\frac{\partial}{\partial x_j}(f)\right)_p\right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)$$

□

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \to V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function. If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transformes from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = \mathrm{d}\varphi_{12}^t \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 2.0.6. By Sylvesters Law of interta, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted with λ_p. If we cannot suppres the function without loss of clearity we will write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

Theorem 2.0.7 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse funktion $f : M \to \mathbb{R}$. There exists a centered chart $\varphi : U \to \mathbb{R}^m$ arround p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2$$

Proof. Compar for example Lemma 3.11 in [BH04]. ☐

hier noch mehr Quellen!

Definition 2.0.8. Let $\varphi_t : M \to M$ be the local one-parameter group of diffeomorphisms generated by $-\mathrm{grad}(f)$, i.e. if $e : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) \circ e = -\mathrm{grad}(f)(\varphi_t(x)) \,, \quad \text{and}$$
$$\varphi_0(x) = x \,.$$

We call that group the **negative gradient flow** of f with respect to the metric g. The integral curve $\gamma_x : (a, b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Next we will deepen our **local** understanding of said vector field. Notice how the Morse lemma is a statement about smooth manifolds no metric needed nor mentioned. In fact there are many Morse charts and we can find one, where the Riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 2.0.9. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the tangent bundle in a point p that come from a basis in said point induced from a chart χ by q_χ. Then the diagramm*

$$\begin{tikzcd} T_pM \arrow[r, "q_\psi|_p"] \arrow[dr, "q_{L\circ\psi}|_p"'] & \mathbb{R}^m \arrow[d, "L"] \\ & \mathbb{R}^m \end{tikzcd}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi\big|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi\circ L}|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x}_0$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}\big|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\big|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
\left[(q_{\psi\circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\Big|_p (f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}\Big|_{\tilde{x}_0} (f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\Big|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\Big|_p (f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[q_\psi\big|_p^{-1}\big(\sum_{i,j} v_i \lambda_{ji} e^j\big)\right](f_p) \\
&= \left[q_\psi\big|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. □

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 2.0.10 (Simultaneous Diagonalisation of Quadratic Forms). *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ arround p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a a diagonal matrix $diag(\mu_1, \ldots \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precice this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_\psi\big|_p \oplus q_\psi\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda G v$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \ldots, \lambda_m$ and corresponding eigenvectors $v_1, \ldots, v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i)\,.$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1|\dots|v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^m z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H(Lz) = z^T L^T H L z \end{aligned}$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive).This is true to sysvesters law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\operatorname{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \operatorname{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0\ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^{l} -y_i^2 + \sum_{i=l+1}^{m} y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p\right)$. By the lemma 2.0.9

$$g(q_{L^{-1}\circ\psi}\big|_p^{-1}, q_{L^{-1}\circ\psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 2.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulars around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 2.0.12 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\mathrm{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\Big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\Big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\mathrm{grad}(f))_\psi(x) := J\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial (q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}$$

Lemma 2.0.13. *Let ψ be a coordinate system around a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\Big|_x, \dots, \frac{\partial}{\partial x_m}\Big|_x\right)$ be the induced basis of T_xM. Then the Jacobian of the gradient reads*

$$J(\mathrm{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij}, \tag{4}$$

where g^{ij} is defines as in definition 1.0.17.

Proof. Let q_ψ denote the coordinate funktion $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\Big|_x, \dots, \frac{\partial}{\partial x_m}\Big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \mathrm{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned}
\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\
&= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j\, \partial x_k}\right]\right)_{ij}.
\end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j\, \partial x_k}\right]\right)_{ij}.$$

□

Now that we now how the gradient locally looks like its reasonable to assume, that the flow locally looks like its exponential:

Lemma 2.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ is a critical point. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t,x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{dt}}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{dt}}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceding lemma now finishes the proof.

□

Hence in a refind Morse chart we have:

Remark 2.0.15. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$
\begin{aligned}
\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \exp(-J(\operatorname{grad}(f))(p)\cdot t) \\
&= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f\circ\psi^{-1}}{\partial x_j \partial x_k}}_{=s_k 2\delta_{jk}}\cdot t\right) \\
&= \operatorname{diag}\left(e^{2\mu_1 t},\dots,e^{2\mu_{\lambda_p} t},-e^{2\mu_{\lambda_{p+1}} t},\dots,-e^{2\mu_m t}\right)
\end{aligned}
$$

In a refined Morse chart we have that the (un-)stable Manifold embedding:

Definition 2.0.16 (Stable and Unstable Manifold)**.** Let $p \in M$ be a critical point of $f : M \to \mathbb{R}$. Then we define:

1. The **stable set** of p as
$$W(\to p) := \{x \in M | \lim_{t\to\infty} \varphi_t(x) = ß\}.$$
That is the points that flow towards p.
2. The **unstable set** of p as
$$W(p \to) := \{x \in M | \lim_{t\to-\infty} \varphi_t(x) = ß\}.$$
That is the points that came from p.

Those sets will turn out to be manifolds, which explains the names **(un-)stable manifolds**.

The main theorem of this chapter will be the:

Theorem 2.0.17 (Stable and unstable manifold theorem)**.** *Let $f : M \to \mathbb{R}$ be a smooth function on a smooth compact Riemannian manifold (M, g). Let p be a critical point of f. Then the tangent space splits as:*

$$T_pM = T_p^s(M) \bigoplus T_p^u M$$

such that the Hessian is positive definite on $T_p^s M$ and negative definite on $T_p^u M$. Moreover, the stable and unstable manifolds are submanifolds and images of smooth embeddings:

$$
\begin{aligned}
E_p^s : T_p^s M &\to W(\to p) \subset M, \\
E_p^u : T_p^u M &\to W(p \to) \subset M.
\end{aligned}
$$

Proof. For the proof compare [BH04]. □

We want to refine the unstable embedding $E_q^u : T_q^u M \to W(q \to)$. For this we will make use of the Grobman-Hartman conjugation theorem:

Theorem 2.0.18 (Hartman-Grobman Theorem)**.** *Let E be an open subset of R^n cobtaining the origin, let $f \in \mathcal{C}$, and let φ_t be a flow given by f. (i.e. $\frac{\partial}{\partial t}(\varphi_t(x)) = f(x)$). Assume that $f(0) = 0$, i.e. that 0 is a fixpoint of φ_t and furthermore assume that fixpoint to be hyperbolic, i.e. Df has no eigenvalue with zero real part. Then there exists a homeomorphism H of an open set U containing the origin onto an open set V containing the origin such that for each $x_0 \in I$, there is an open Interval $I_0 \subseteq \mathbb{R}$ containing zeri such that for all $x_0 \in U$ and $t \in I_0$ the diagram*

$$\begin{tikzcd} U \arrow[r, "H"] \arrow[d, "\varphi_t"] & V \arrow[d, "e^{At}"] \\ U \arrow[r, "H"] & V \end{tikzcd}$$

commutes ,meaning tha locally the flow is a topological conjugate to its linearisation.

Proof. Compare chapter 2.8 in [Per01]. □

Theorem 2.0.19 (Refined Stable Embdeding)**.** *In the setting of the stable manifold theorem, we can refine the embedding*

$$E_q^u : T_q^u M \to W(q \to)\,,$$

such that it resembles the flow, meaning that for $x = E_p^s(v)$ the flow is given by

$$\varphi_t(x) = E_p^s(A_t v)\,,$$

where A_t is a linear map defined as: $A_t = B_{\mathfrak{b}} \circ \operatorname{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}\right) \circ B_{\mathfrak{b}}^{-1}$, where $\mathfrak{b}$ is the bases of $T_q^u M$ that comes from the basis of $T_q M$ which is induced by a refined Morse chart. The diagonal matrix is the stable part of the jacobean of the flow in said refined Morse chart ψ.

Proof. In the proof of the unstable manifold theorem we started with a local map $\rho' : U \to V$, where $U \subseteq W(q \to)$ is an neighbourhood of q and $V \supseteq T_q^u$. (This was a refined Morse chart together with a projection once realized that the stable part gets mapped to the graph of a smooth funktion). Now we define

$$\begin{aligned}\theta_t : V &\to V\\ v &\mapsto \rho' \circ \varphi_t \circ \rho'^{-1}(v)\end{aligned}$$

for all t such that the map is well defined. Now with a basis $\mathfrak{b}_p^s$ of $T_p^s M$ we can push these considerations into $\mathbb{R}^n$ and by abuse of notation assume that ρ maps to $\mathbb{R}^n$ and hence θ is a flow arround the origin of $\mathbb{R}^n$. Now by the Hartman-Grobman Theorem we find a homeomorphism H arround the origin, such that $H \circ \theta_t = e^{At} \circ H$, where $A = \mathrm{d}\theta_t =$ To calculate said differential we look into the map ρ' again. For this we give names to the funktions:

- $\psi : U \to V$ is the refined morse chart arround q.
- $g : T_p^s M \to T_p^u M$ is the map that defines the image of the stable part under ψ to be a graph. I.e. $\psi\left(W(\to p)\right) = \Gamma_g$. We know that $g(0) = 0$ and $\mathrm{d}g\big|_0 = 0$.
- $\pi : \Gamma_g \to T_p^s M$ is the projection. Its inverse is obviously $\mathrm{id} \times g$.

We know from the local stable manifold theorem, that $\mathrm{d}g\big|_0 = 0$. With this we can calculate:

$$\begin{aligned}\mathrm{d}\theta_t\big|_0 &= \mathrm{d}\left(\pi \circ \psi \circ \varphi_t \circ \psi^{-1} \circ \pi^{-1}\right)\big|_0\\ &= \mathrm{d}\pi\big|_{(0,0)} \circ \mathrm{d}\psi \circ \varphi_t \circ \psi^{-1}\big|_{0,0} \circ \mathrm{d}\,\mathrm{id} \times g\big|_0\\ &= \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right) \circ \mathrm{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}, e^{2\mu_{\lambda_q+1} t}, \dots, e^{2\mu_m t}\right) \circ \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right)^T\\ &= \mathrm{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}\right)\end{aligned}$$

Now define for a small neighbourhood $V \subseteq \mathbb{R}^{m-\lambda_p}$ the map

$$\rho := \rho' \circ H : V \to U\,.$$

U is choosen such that the map is onto. Then we have that $\varphi_t \circ \rho^{-1} = \rho^{-1} \circ \mathrm{e}^{At}$, in a that small neighbourhood of p. Now finally we can extend this property to be true for the whole stable set: For this we define the embedding

$$\begin{aligned}E_p^s : T_p^s M &\to W(\to p)\\ x &\mapsto \varphi_{-n} \circ \rho^{-1} \circ A_n(x) \qquad \text{where } n \in \mathbb{N} \text{ is big enough such that } A_n(x) \in V\,.\end{aligned}$$

This map is in fact well defined since

$$\begin{aligned}\varphi_n \circ \rho^{-1} \circ A_n(x) &= \varphi_{-n} \circ \varphi_{-1} \circ \varphi_1 \circ \rho^{-1} \circ A_n(x)\\ &= \varphi_{-n} \circ \varphi_{-1} \circ \rho^{-1} \circ A_{n+1}(x)\\ &= \varphi_{-(n+1)} \circ \rho^{-1} \circ A_{n+1}(x)\end{aligned}$$

And obviously the flow compatibility that is given locally by ρ is now true globally: Let $x \in W(\to p)$ and assume $x = E_p^s(y)$. Then:

$$\begin{aligned}\varphi_t(x) = \varphi_t(E_p^s(y)) &= \varphi_t \circ \varphi_{-n} \circ \rho^{-1} \circ A_n(y) \\ &= \varphi_{-n} \circ \varphi_t \circ \rho^{-1} \circ A_n(y) \\ &= \varphi_{-n} \circ \circ A_n(A_t(y))\end{aligned}$$

□

Now we want to understand the flow in an unstable set and analyze two maps:

Remark 2.0.20. Let q be a critical point and $c < f(q)$ such that no other critical point lies in $f{-}1[c, f(q)]$ then, the map

$$\begin{aligned}\tau : W(q \to) \smallsetminus \{q\} &\to \mathbb{R} \\ x &\mapsto \tau(x) \text{ such that } f \circ \varphi_{\tau(x)}(x) = c\end{aligned}$$

is smooth.

Proof. We know that such a map exists and is well defined since every flow line passes through $f^{-1}(c)$. Now let $x \in W(q \to) \smallsetminus \{q\}$, V be a neighbourhood of x and U be a neighbourhood of $\varphi_{\tau(x)}(x)$. Define the map

$$\begin{aligned}G : V \times I &\to \mathbb{R} \\ (x, t) &\mapsto f \circ (\varphi_t(x)) \,.\end{aligned}$$

Now the partial derivative reads

$$\frac{\partial}{\partial t} G\big|x, t = \mathrm{d}f\big|_{\varphi_t(x)} \circ \frac{\partial}{\partial t}\varphi_t(x)\big|_t = \mathrm{d}f\big|_{\varphi_t(x)} \circ \left(-\mathrm{grad}_{\varphi_t(x)}(f)\right) .$$

For $\varphi_{\tau(x)}$ we have that $T_{\varphi_{\tau(x)}(x)} f^{-1}(c) = \ker \mathrm{d}f\big|_{\varphi_{\tau(x)}(x)}$. Since the flow line is transversal to the level set or said differently, the gradient is orthogonal to level sets, we have that the latter does not live in the kernal and hence

$$\frac{\partial}{\partial t} G\big|x, \tau(x) = \mathrm{d}f\big|_{\varphi_{\tau(x)}(x)} \left(-\mathrm{grad}_{\varphi_{\tau(x)}(x)}(f)\right) \neq 0 \,.$$

This lets us use the implicit function theorem to deduce that the level set $G = c$ is locally the graph of a function $\tau : U_x \to \mathbb{R}$ where U_x is a neighbourhood of x. Hence the function τ is globally smooth since this argument can be done everywhere. □

Remark 2.0.21. Now we care for the differential of τ in a point $x_j \in W(q \to) \cap f^{-1}(c)$. For this we choose a chart of $W(q \to)$ arround x_j such that the induced bases of $T_{x_j} W(q \to) = \langle -\mathrm{grad}_{x_j}(f) \rangle \oplus T_{x_j} V_j$ that respects the splitting. for example $\frac{\partial}{\partial x^1}\big|_{x_j} =$

$-\mathrm{grad}_{x_j}(f)$ and $\frac{\partial}{\partial x^i}\big|_{x_j} \in T_{x_j}V_j$ for i bigger than one. With this we can start our calculations: Notice how $x \mapsto f \circ \varphi_{\tau(x)}(x)$ is a constant function and hence its differential is 0.

$$\begin{aligned} 0 = \mathrm{d}_x\left(f \circ \varphi_\tau\right) &= \overbrace{\mathrm{d}f}^{\text{1-form}}\ (\mathrm{d}\varphi_\tau) = \mathrm{d}f\left(\mathrm{d}_t\varphi_t \circ \mathrm{d}_x\tau(x) + \mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right) \\ &= \mathrm{d}f\left(\mathrm{d}_t\varphi_t \circ \mathrm{d}_x\tau(x)\right) + \mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right) \\ &= \mathrm{d}_t\left(f \circ \varphi_t\right) \circ \mathrm{d}_x\tau(x) \circ + \mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right) \end{aligned}$$

Hence we know that

$$\mathrm{d}_x\tau = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right)}{\mathrm{d}_t\left(f \circ \varphi_t\right)} = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right)}{-\left\|\mathrm{grad}(f)^2\right\|} \tag{5}$$

Now we can calculate in x_j where $\tau(x_j) = 0$

$$\mathrm{d}_x\tau\big|_{x_j}\frac{\partial}{\partial x^i}\Big|_{x_j} = \frac{\mathrm{d}f\big|_{x_j}\frac{\partial}{\partial x^i}\big|_{x_j}}{\left\|\mathrm{grad}(f)^2\right\|} = \frac{g\left(\mathrm{grad}_{x_i}(f), \frac{\partial}{\partial x^1}\big|_{x_j}\right)}{\left\|\mathrm{grad}(f)^2\right\|} = \begin{cases} -1 & \text{if } i = 1 \\ 0 & \text{else.} \end{cases}$$

This result can be extended to also work outside of x_j as follows: Notice the identity for all $s \in \mathbb{R}$ and $x \in W(q \to) \setminus \{q\}$:

$$\tau(\varphi_s(x)) = \tau(x) - s\,.$$

Differentiating this in the variable s and in for s' yealds

$$\mathrm{d}\tau\big|_{\varphi_{s'}(x)} \circ \underbrace{\mathrm{d}\varphi_s(x)\big|_{s=s'}}_{=-\mathrm{grad}_{\varphi_s(x)}(f)} \;=\; \mathrm{d}\tau(\varphi_{s'}(x))\big|_{s'} \;=\; \mathrm{d}\tau(x) - s\big|_{s'} = -1\,.$$

And hence in $s' = 0$ we have for all $x \in W(q \to) \setminus \{q\}$:

$$\mathrm{d}\tau\big|_x\left(-\mathrm{grad}(x)(f)\right) = -1\,.$$

We can furthermore define a basis of $T_yW(q \to)$ with $\mathfrak{b}_1 := -\mathrm{grad}_y(f)$ and $\mathfrak{b}_i = \mathrm{d}\varphi_{\tau(y)}\big|_y^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j}V_j$. Hence w_i is orthogonal on $-\mathrm{grad}_{x_j}(f)$ for all $i > 1$. Hence by the equation (5) we can caluclate:

$$\mathrm{d}_x\tau(\mathfrak{b}_i) = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}(\mathfrak{b}_i)\right)}{-\left\|\mathrm{grad}(f)^2\right\|} = -\frac{\mathrm{d}f\left(w_i\right)}{-\left\|\mathrm{grad}(f)^2\right\|} = 0 \quad \text{for all } i > 1\,.$$

Definition 2.0.22. Let q be a critical point and c such that $c < f(q)$ and $f{-}1[c, f(q)] = q$. Then we define the map

$$\begin{aligned} \rho : W(q \to) \setminus \{q\} &\to W(q \to) \cap f^{-1}(c) \\ x &\mapsto \varphi_{\tau(x)}(x)\,. \end{aligned}$$

Lemma 2.0.23. *Assume that* $\mathfrak{d} = (\mathfrak{d}_i)$ *is a basis of* $T_q^u M$*. Let* $y := E_q^{u-1}(x_j)$ *and* $y' \in (\rho \circ E_q^u)^{-1}(x_j)$*. Then the map*

$$\mathrm{d}E_q^u\big|_y^{-1} \circ \mathrm{d}\rho \circ \mathrm{d}E_q^u\big|_{y'} \circ Q_{y'} \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q} \to T_y T_q^u M$$

can be written as

$$\begin{aligned} \pi_{-\mathrm{grad}_{x_j}(f)^\top} \circ (B_{\mathfrak{b}_{x_j}^q}) : \mathbb{R}^{\lambda_q} &\to T_{x_j} V_j \\ e_i &\mapsto \begin{cases} 0 & \textit{for } i = 1 \\ (\mathfrak{b}_{x_j}^q)_i & \textit{for } i \neq 1\,. \end{cases} \end{aligned}$$

where $(\mathfrak{b}_{x_j}^q)_j$ *is basis of* $T_{x_j} W(q \to)$ *that is induced from* $\mathfrak{d}$ *and* $(\mathfrak{b}_{x_j}^q)_i = -\mathrm{grad}_{x_j}(f)$*. In other words, this is a basis compatible with the one used in the counting of flow lines in the Morse boundary operator.*

Proof Strategy:
To proove this, we first remined ourself how the basis $(\mathfrak{b}_{x_j}^q)_{j>1}$ is constructed. It is a basis that when extendet with $-\mathrm{grad}_{x_j}(f)$ as the first vector is compatible with $\mathfrak{d}$. Hence, we can construct it from $\mathfrak{d}$, by assuming that $\mathrm{d}E_q^u\big|_y \circ Q_y \circ \mathfrak{d}_1 = -\mathrm{grad}_y(f)$. Then, if we push $\mathfrak{d}$ to $T_{x_j} W(q \to)$ via $\mathrm{d}E_q^u\big|_y \circ Q_y$ and delete the first vector, we get the Basis we wanted. So what we need to check in order to proof this lemma is, if the maps

$$\begin{aligned} \mathrm{d}E_q^u\big|_y \circ Q_y \circ B_{\mathfrak{d}} :& \mathbb{R}^{\lambda_q - 1} \to \left(-\mathrm{grad}_{x_j}(f)\right)^\top \\ \mathrm{d}E_q^u\big|_y^{-1} \circ \mathrm{d}\rho \circ \mathrm{d}E_q^u\big|_{y'} \circ Q_{y'} \circ B_{\mathfrak{d}} :& \mathbb{R}^{\lambda_q - 1} \to \left(-\mathrm{grad}_{x_j}(f)\right)^\top \end{aligned}$$

induce the same orientation when restricted to the same domain such that they become vector space isomorphisms. To proof this we will work from the bottom up: We adapt the basis $\mathfrak{d}$ to work with the lower map and show, later, that the adapted basis fits the requirements above:

Proof. We start by defining the space $\left(\tau \circ E_q^u\right)^{-1}(\tilde{t})$, where $\tilde{t} := \tau \circ E_q^u(y')$. As a level set, this is a submanifold of $T_q^u M$ and its tangend space in y' is given by the basis $E_q^u\big|_{y'}^{-1}(\mathfrak{b}_{\mathrm{i}})$ for $i > 1$, where $\mathfrak{b}_i = \mathrm{d}\varphi_{\tilde{t}}\big|_{y'}^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j} V_j$. Now we have the equality of maps:

$$A_{\tilde{t}}\big|_{(\tau \circ E_q^u)^{-1}(\tilde{t})} = \left(E_q^{u-1} \circ \rho \circ E_q^u\right)\big|_{(\tau \circ E_q^u)^{-1}(\tilde{t})}$$

Hence for a basis $\mathfrak{d}_i$ of $T_q^u M$ such that $\mathrm{d}E_q^u\big|_{y'} \circ Q_{y'}(\mathfrak{d}_{\mathrm{i}})$ is $-\mathrm{grad}_{y'}(f)$ for $i = 1$ and $\mathfrak{b}_{\mathrm{i}}$ else, we have:

$$\mathrm{d}\left(E_q^{u-1}\circ\rho\circ E_q^u\right)\Big|_{y'}\circ Q_{y'}(\mathfrak{d}_i) = Q_y\circ A_{\tilde t}(\mathfrak{d}_i).$$

For $i = 1$, the left side maps to 0, and the right side:

$$\begin{aligned}
Q_y\circ A_{\tilde t}(\mathfrak{d}_1) &= \mathrm{d}A_{\tilde t}\big|_{y'}\circ Q_{y'}(\mathfrak{d}_1)\\
&= \mathrm{d}A_{\tilde t}\big|_{y'}\circ \mathrm{d}E_q^u\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f))\\
&= \mathrm{d}A_{\tilde t}\circ \mathrm{d}E_q^u\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f))\\
&= \mathrm{d}(E_q^u)^{-1}\circ\varphi_{\tilde t}\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f))\\
&= \mathrm{d}E_q^u\big|_{x_j}^{-1}\circ \mathrm{d}\varphi_{\tilde t}\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f))\\
&= \mathrm{d}E_q^u\big|_{x_j}^{-1}\left(-\mathrm{grad}_{x_j}(f)\right)
\end{aligned}$$

The last equation is the general fact prooven in remark 2.0.15, that for fixed t the identity holds:

$$\mathrm{d}_x\varphi_t\big|_y(-\mathrm{grad}_y(f)) = -\mathrm{grad}_{\varphi_t(y)}(f)$$

Hence if π denotes the projection onto the complement of $\mathrm{d}E_q^u\big|_{x_j}^{-1}\left(-\mathrm{grad}_{x_j}(f)\right)$ we have an equality of maps

$$\mathrm{d}\left(E_q^{u-1}\circ\rho\circ E_q^u\right)\Big|_{y'}\circ Q_{y'} = \pi\circ Q_y\circ A_{\tilde t}.$$

Or by the above calculation, we can throw out π, if we exclude $\mathfrak{d}_1$. Furthermore, the equation shows that our requirement $\mathrm{d}E_q^u\big|_{y'}\circ Q_{y'}(\mathfrak{d}_1) = -\mathrm{grad}_{y'}(f)$ implies that $\mathfrak{d}$ is compatible with a basis where $\mathrm{d}E_q^u\big|_y\circ Q_y\circ\mathfrak{d}_1 = -\mathrm{grad}_y(f)$, since the sign of the determinant of $A_{\tilde t}$ is always positive. This concludes the prove! $\square$

Definition 2.0.24 (Morse Homology)**.** Let $f: M\to\mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M,g). Furthermore let $\mathfrak{o}$ be an choice of orientations for all unstable Tangendspaces (T_q^uM for all $q\in\mathrm{Crit}(f)$). The tuple $\mathfrak{A} := (g,f,\mathfrak{o})$ is a **Morse datum**. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ):

$$C_k(M,\mathfrak{A},\Lambda) := \bigoplus_{\mathrm{Crit}(f)}\Lambda$$

To define a boundary operator between the chain groups on the generators we count flow lines between them. Lets say that p is a critical point of index k and q one of index $k+1$. Then we have the set of flow lines from q to p denoted by $\mathcal{M}(q\to p)$. For each $\gamma\in\mathcal{M}(q\to p)$ we assign a sign $n_\gamma\in\{\pm1\}$. Let $x\in\gamma$. Then the tangent space in x splits $T_xW(q\to) = N_x(\gamma)\oplus\langle -\mathrm{grad}(f)(x)\rangle$. Furthermore, we can parallel transport $N_x(\gamma)$ to T_p^uM. Hence, the orientation in T_q^uM can be compared to the one in T_p^uM and the sign n_γ can be chosen accordingly.The **boundary operator** on a generator is defined as:

$$\partial_{k+1}(q) := \sum_{p\in\mathrm{Crit}_k(f)}\ \sum_{\gamma\in\mathcal{M}(q\to p)} n_\gamma$$

This gives rise to a chain complex and hence gives rise to a homology theory, and a cohomology theory by applying the hom functor to the chain complex.

Definition 2.0.25 (Orientation and Exact Sequences)**.** As stated before, an orientation of a Vector space V of dimension m is given by an ordered basis $\mathfrak{b} = (b_1, \dots b_m)$ of V and hence the produkt $b_1 \wedge \dots \wedge b_m$ is an element of the maximal exterior product $\det(V) = \Lambda^m V$. Two ordered bases define different elements in $\det(V)$ where there difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence an orientation is a choice of one connected component of $\det(V) \smallsetminus \{0\}$. Now a short exact sequence of vector spaces

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

induces a canonical isomorphism

$$\begin{aligned} \det(A) \otimes \det(C) &\to \det(B) \\ (a_1 \wedge \dots \wedge a_l) \otimes (c_1 \wedge \dots \wedge c_k) &\mapsto (f(a_1) \wedge \dots f(a_l) \wedge \tilde{c_1} \wedge \dots \wedge \tilde{c_k}) \,, \end{aligned}$$

where $\tilde{c_j} \in g^{-1}(c_j)$. The well definition of this map (invariance under choosen lift of c) can be calculated, since two different choices differ in an element in the kernal of g which equals the image of f and hence the different choices vanisch in the wedge produkt. By this isomorphism we get a orientation induced in the middle from the outside.

Remark 2.0.26 (The Counting of Flow Lines)**.** There are several ways to formulate the counting of flow lines that happens in the boundary operator.Let $\gamma \in \mathcal{M}(q \to p)$ and $x \in \gamma$ Instead of using the parallel transport, we can introduce two different ordered bases in $T_x W(q \to)$. We use the following notation: Let $\gamma \in \mathcal{M}(q \to p)$ and $x \in \gamma$. For this we define $N_x^{W(q\to)}\gamma$ be the normal bundle of γ as a submanifold of $W(q \to)$. Since γ is contractible we can trivialize said normal bundle to get an bundle isomorphism:

$$\nu : N^{W(q\to)}\gamma \to \gamma \times T_p^u M \,.$$

Let $\nu\big|_x$ denote the isomorphism $N_x^{W(q\to)}\gamma \to T_p^u M$. An orientation of $T_p^u M$ induces then one on $N_x^{W(q\to)}\gamma$ and if we define the bases $(-\mathrm{grad}_x(f))$ of its own span to be positive, we get the exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \langle -\mathrm{grad}_x(f) \rangle \arrow[r] & T_x W(q \to) \arrow[r] & N_x^{W(q\to)}\gamma \arrow[r] & 0 \\ & & & T_p^u M \arrow[u, "(\nu_x)^{-1}"'] & \end{tikzcd}$$

That introduces an orientation in the middle that can be represented by the basis $(-\mathrm{grad}_x(f), i \circ \nu|_x)(\mathfrak{b}^p)$, where i is the inclusion $N_x^{W(q\to)}\gamma \to T_x W(q \to)$ and $(\mathfrak{b}^p)$ is a ordered basis ot $T_p^u M$. Another way to get an orientation there is to trivialize the bundle $TW(q \to)$ via the isomorphism:

$$\mu : TW(q \to) \to W(q \to) \times T_q^u M \,; \xi_y \mapsto \left(y, Q^{-1}_{E_q^{u-1}(y)} \circ \mathrm{d}E_q^{u-1}\big|_y (\xi_y) \right) .$$

This trivialization together with an orientation on $T_q^u M$ orients the middle in the above exact sequence. Now the deciding factor that gives γ its sign is weather the two orientations agree. This can by checked by calculating the determinant of a mcertain map. To formulate the map somewhat readible, we assume that $\mathfrak{b}^q$ is a basis of $T_q^u M$ such that for $y = E_q^{u-1}(x)$ we have $\mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q) = -\mathrm{grad}_x(f)$ for $i = 1$, and for all $i > 1$ the image lives in the orthogonal complement of $-\mathrm{grad}_x(f)$ and hence in the domain of ν_x. Then we can define the map

$$\begin{aligned} B_{\mathfrak{b}^p}^{-1} \circ \nu_x \circ \mu_x^{-1} \circ B_q : \mathbb{R}^{\lambda_q} \supseteq \mathbb{R}^{\lambda_q - 1} = \langle e_2, \dots e_{\lambda_q} \rangle \to \mathbb{R}^{\lambda_q - 1} \\ e_i \mapsto B_{\mathfrak{b}^p}^{-1} \circ \nu_x \circ \mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q)\,, \end{aligned}$$

and the determinant of said map gives us n_γ.

Remark 2.0.27. Notice how we already saw a different way to calculate the sign by the map from lemma 2.0.23.

Definition 2.0.28. Let $(M, \mathfrak{A}, \Lambda)$ and $(M', \mathfrak{A}', \Lambda)$ be two Morse data with $\mathfrak{A} = (g, f, \mathfrak{o})$ and $\mathfrak{A}' = (g', f', \mathfrak{o}')$. Then on $M \times M'$ we define the Morse function $f + f'$ and the metric $g \oplus g'$. Then $\mathrm{d}d_{(x,x')}(f + f') = \mathrm{d}_x(f) + \mathrm{d}'_x(f')$ and hence the critical points of $f + f'$ are pairs (p, p') such that p is critical for f and p' is critical for f'. By corollary 2.0.5 we conclude that the hessian splits into block diagonal form and hence the property of non degenerayc follows. Since we use the product metric $g \oplus g'$ we also get a splitting gradient of $f + f'$:

$$\begin{aligned} \mathrm{d}_{(x,x')}(f + f')(V, V') &= \mathrm{d}_x(f)(V) + \mathrm{d}'_x(f')(V') \\ &= g(\mathrm{grad}_x(f), V) + g'(\mathrm{grad}'_x(f'), V') \\ &= (g \oplus g')\big(\underbrace{(\mathrm{grad}_x(f), \mathrm{grad}'_x(f'))}_{\mathrm{grad}_{(x,x')}(f+f')}, (V, V')\big)\,. \end{aligned}$$

Hence the flow also splits and thereby the (un-)stable manifolds split Hence we can orient all unstable manifolds via that splitting whcih gives os $\mathfrak{o} \oplus \mathfrak{o}'$. Finally transversal intersection is a property that carries over to products and hence the Morse datum $(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda)$ with $\mathfrak{A} \times \mathfrak{A}' := (g \oplus g', f + f', \mathfrak{o} \oplus \mathfrak{o}')$ really gives a valid Morse datum.

Definition 2.0.29 (Morse Cohomology)**.** Let G be a commutative ring with one. To get the cohomology we have to dualice the chain complex. Let $\mathrm{Crit}_k(f) = \{p_1, \dots p_l\}$ be a basis of the k-th chain group $C_k(M, \mathfrak{A}, G)$. Then the functionals

$$\eta_i : C_k(M, \mathfrak{A}, G) \to G; p_j \mapsto \begin{cases} 1 & \text{if } \text{j=i} \\ 0 & \text{else.} \end{cases}$$

are a basis of the dial group $C^k(M, \mathfrak{A}, G)$. The coboundary map is easy to obtain too. Let q_j for $j = 1, \dots, r$ be the critical points with index $k+1$ and let ξ_i be the corresponding forms. Then let $n_f(q_j, p_i)$ be the number of flow lines form q_j to p_i counted with the sign. Then the coboundary map is given as:

$$\begin{aligned}\partial^k : C^k(M, \mathfrak{A}, G) &\to C^{k+1}(M, \mathfrak{A}, G)\\ \eta_i &\mapsto \partial^k \eta_i = \eta_i \circ \partial_{k+1} = \sum_j n_j(q_j, p_i)\xi_j\,.\end{aligned}$$

Corollary 2.0.30. If M is an oriented manifold, orientations from the unstable tanged space in critical points $T_q^u M$ induce Orientations in the stable tangend space since

$$T_q M = T_q^u M \otimes T_q^s M\,.$$

Here i write a little bit about the Morse cohomology in german

Theorem 2.0.31 (Morse Cohomology)**.** *Um die Morse Kohomologie zu erhalten, dualizieren wir den Kettenkomplex für einen Ring G mit 1. Sei* $\mathrm{Crit}_k = p_1, \ldots p_l$ *eine Basis der* $k-$*ten Kettengruppe. Für*

$$\begin{aligned}\eta_i : C_k &\to G\\ p_j &\mapsto \begin{cases} 0 & \text{if } j = i\\ 1 & \text{else}\end{cases}\end{aligned}$$

bekommen wir eine Basis $(\eta_1, \ldots, \eta_l)$ *der Cocetten Gruppe. Die Randabbilsung ist nun der spannende Teil. Seien* $(q_1, \ldots, q_r)$ *kritische Punkte von Index* $k+1$ *und* $(\xi_j)_{j=1,\ldots,r}$ *die resultierende Basis von*C^{k+1}*. Dann*

$$\begin{aligned}\mathrm{d}_k : C^k &\to C^{k+1}\\ \eta_i &\mapsto \mathrm{d}\eta_i = \eta_i \circ \partial_{k+1} = \sum_j n_f(q_j, p_i)\xi_j.\end{aligned}$$

Hier beschreibt $n_f(q,p)$ *das zählen von Flusslionien von* q *nach* p *mittels* f*.Indem wir kritische Punkte und die formen identifizieren erhalten wird das folgende:*

$$\begin{aligned}C^k &= \bigoplus_{q \in \mathrm{Crit}_k} \mathbb{Z} = C_k.\\ \mathrm{d}: \ C^k &\to C^{k+1}\\ q &\mapsto \sum_{p \in \mathrm{Crit}_{k+1}} n_f(q,p) = \sum_{p \in \mathrm{Crit}_{k+1}} n_{-f}(p,q) = \partial_{-f}.\end{aligned}$$

Hier bezeichnet ∂_{-f} *den Randoperator* $C(M, -f, g)$.

Theorem 2.0.32 (Poincaré Dualität)**.** *Schnell sehen wir, dass*

$$C^k(M, f) = C_{m-k}(M, -f)$$

damit erhalten wir:

$$H^k(M) \cong H^k(M, f) = \ker(\partial_{-f}) \big/ \mathrm{im}\partial_{-f} = H_{m-k}(M, -f) \cong H_k(M)$$

3 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

3.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 3.1.1 (Compact invariant isolated subset)**.** For a flow $\varphi_t : M \to M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N | \ \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t \in \mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 3.1.2 (Index pairs)**.** Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of** S, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (N \overset{\circ}{\setminus} L)$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in** N. Here $\varphi_{[0,t]}(x) := \{\varphi_{\tilde{t}}(x) | \tilde{t} \in [0, t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S. Due to the third property we call L the **exit set**.

Definition 3.1.3 (Regular index pairs)**.** We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 3.1.4 (Homotopy equivalence of index pairs)**.** *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S. Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces. This homotopie is canonical in transitive up to homotopy.*

Definition 3.1.5 (Connected Simple System)**.** A **connected simple system** consists of a cllection I_0 of pointed spaces along with a collection I_m of homotopy classes of maps between these spaces such that

(i) $\hom(X,Y) := \{[f] \in [X:Y] : [f] \in I_m\}$ consists of exactly one element for each pair of spaces $(X,Y) \in I_0^2$, where $[X:Y]$ denotes the set of homotopy classes of maps between X and Y.

(ii) If $X,Y,Z \in I_0, [f] \in \hom(X,Y)$, and $g \in \hom(Y,Z)$, then $[g \circ f] \in \hom(X,Z)$.

(iii) $\hom(X,Y) = \{[\mathrm{id}_X]\}$ for all $X \in I_0$.

Remark 3.1.6. This definitions comes from Salamon in [Sal85].

Remark 3.1.7. Let S be a isolated, compact, invariant subset of M with respect to the Morse flow φ. Then the collections

$$I_0 := \{N/L : (N,L) \text{ is index pair for } S\}$$
$$I_m := \{[f] : f \text{ is the homotopy equivalence of index pairs from the proof of theorem 3.1.4}\},$$

make up a connected simple system. We call this $\mathcal{IP}_S$.

Definition 3.1.8 (Morphisms of Connected Simple Systems)**.** A **morphism** $\Phi : I \to Y$ between two connected simple systems $I = (I_0, I_m)$ and $J = ()J_0, J_m)$ is a collection of homotopy classes of maps between spaces io I_0 and J_0 such that

(i) For every $X \in I_0$ and $Y \in J_0$ the set $\Phi(X,Y) := \{[\varphi] \in [X:Y] : [\varphi] \in \Phi\}$ consits of exactly one element.

(ii) If $X, \bar{X} \in I_0$ and $Y, \bar{Y} \in J_0$ and if $[\varphi] \in \Phi[X,Y]$,$[f] \in \hom[\bar{X},X]$,$g \in \hom(Y,\bar{Y})$, then $[g \circ \varphi \circ f] \in \Phi(\bar{X},\bar{Y}$

Remark 3.1.9. By the second property of a morphism between connected somple systems I and J we can induce such a single map $\varphi : X \to Y$ where $X \in I_0$ and $Y \in Y_0$.

Remark 3.1.10 (Critical points as isolated compact invariant subsets)**.** Let $f : M \to \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M,g). Let $q \in \mathrm{Crit}_k(f)$. Around q we have Morse charts $\varphi : U \to T_qM$ (after identifying q_i and ∂q_i) where $\varphi(W(q \to) \cap U) \subseteq T_q^uM$ and $\varphi(W(\to q) \cap U) \subseteq T_q^sM$. With this we can define the balls:

$$\begin{aligned} D_\varepsilon^s &= \{v \in T_q^sM | \,\|v\| \le \varepsilon\}, \\ D_\varepsilon^u &= \{v \in T_q^uM | \,\|v\| \le \varepsilon\}. \end{aligned}$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Now we know the K-group of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i-k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \tag{6}$$

Notice that for $k = 0$ we need to consider $\partial D^k = \varnothing$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k:

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{7}$$

Or using the language of connected simple systems, we have the identification

define what K(IP) should mean

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(\mathcal{IP}_q)). \tag{8}$$

Definition 3.1.11 (Canonical Isomorphism Class)**.** A **canonical isomorphism class** of rings (or other algebraic structures) is a set of rings R_0 together with a set of ring homomorphisms R_m such that

(i) For $R, L \in R_0$ the set $\hom(X, Y) := \{f : X \to Y : f \in R_m\}$ consists of exactly one element.

(ii) If $R, L, S \in R_0$ and $f \in \hom(R, L)$ and $g \in \hom(L, S)$, then $g \circ f \in \hom(X, Z)$.

(iii) $\hom(X, X) = \{\mathrm{id}_x\}$

A **morphism** between two canonical isommorphism classes of rings $\Psi : (R_0, R_m) \to (S_0, S_m)$ is a collection of homomorphisms such that

1. For any $R \in R_0$ and $S \in S_0$ the set $\Psi(R, S) := \{f : R \to S : f \in \Psi\}$ contains exactly one element.

2. If $R, \bar{R} \in R_0$ and $S, \bar{S} \in S_0$, $f \in \hom(\bar{R}, R)$, $g \in \hom(S, \bar{S})$ and $\psi \in \Psi(R, S)$, then $g \circ \psi \circ g \in \Psi(\bar{R}, \bar{S})$.

Remark 3.1.12. Again the second property lets us induce a morphism between two canonical isommorphism classes of rings $\Psi : (R_0, R_m) \to (S_0, S_m)$ by a single isomorphism

$$\psi : R \to S$$

for $R \in R_0$ and $S \in S_0$.

Remark 3.1.13. We call the category of simple connected systems **SCS** and the categorie of canonical isomorphism classes of rings **CICR**

Definition 3.1.14. The classical complex topological K-functors $\tilde{K}^i$ define a family of kontravariant funtors

$$\mathbf{SCS} \to \mathbf{CICR}$$

that is fiven on the objects as follows: For (I_0, I_m) being a family of pointed topological spaces, we get a new family of rings as:

$$R_0 := \{\tilde{K}^i(X) : X \in I_0\},$$

and a family of ringhomomorphisms

$$R_m := \{\tilde{K}^i(f) : [f] \in I_m\}.$$

Obviously, the tuple (R_0, R_m) is an object in **CICR**. Now a morphism in $\Phi : (I_0, I_m) \to (J_0, J_m)$ in **SCS** gifes rise to a collecting of ringisomorphisms

$$\Psi := \{\tilde{K}^i(\varphi) : [\varphi] \in \Phi\}.$$

By the homotopy property of reduced K-theory, this becomes a morphism in **CICR**.

Example 3.1.15. Any pointed topological space X defines a simple connected system $(\{X\}, [\mathrm{id}_x])$. Similar, any ring R defines a caninical isomorphism class of rings $(\{R\}, \{\mathrm{id}_R\})$.

Definition 3.1.16. Let (M, f, g) be a smooth manifold M together with a Morse Smale funtion f. Let $S \subseteq M$ be a compact invariant subset. Then the index pairs of S together with the canonical isomorphisms between them define a simple conected system which we will denote with $\mathcal{IP}_S$.

Remark 3.1.17. Let $\underline{\mathbb{Z}}$ be the canonical equivalence class of rings consiting only of the ring $\mathbb{Z}$, and let p be a critical point of index λ_p in M. Then remark 3.1.10 gives us the existence of a morphism for all $i \in \mathbb{N}$:

$$\tilde{K}^{\lambda_p + 2i}(\mathcal{IP}_p) \to \underline{Z}.$$

The next remark will ehlp is to get a canonical isomorphism.

Remark 3.1.18. Furthermore, this isomorphism

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{9}$$

can be made canonically, using orientations: Let q be a critical point in index λ_q. Let $\mathfrak{b}$ be a ordered basis of $T_q^u M$. Let ψ be a refined Morse chart arround q, such that the induced basis in $T_q M = T_q^u M \oplus T_q^u M$ projects onto a a basis of the first factor $T_q^u M$ that is compatible with $\mathfrak{b}$. Using this Morse chart do define the generator of the regular index pair of q gives a unique generator.

Proof. Let ψ_1 and ψ_2 be two such Morse charts. Notice that they only differ by order of the directions and their signs, meaning that for all i there is a j such that $\psi_1^{-1}(e_i) = \pm\psi_2 - 1(e_j)$. Hence, viewed as maps into the tangent space we have that $\psi_1(N_p, L_q) = \psi_1(N_p, L_q) \subseteq T_q M$. We define $B_\varepsilon^{\lambda_q}(0) \subseteq \mathbb{R}^{\lambda_q}$ to be the sphere. The generator of $\psi_1(N_p, L_q)$ comes as the pullback of the canonical bott generator of $K^i(B_\varepsilon^{\lambda_q}(0), \partial B_\varepsilon^{\lambda_q}(0))$ via the basis map after the collapse map $c : (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Let α be a generator of $\psi_1(N_p, L_q)$ induced from the basis $\mathfrak{b}_1$ that comes from ψ_1. I.e.

$$\alpha = (B_{\mathfrak{b}_1} \circ c)^* (\beta)$$

By assumption, the map $B_{\mathfrak{b}_1} \circ B_{\mathfrak{b}}^{-1}$ has determinant one. Hence it is homotopic to the identity, and this homotopy can be reduced to a well defined homotopy on the quotient space $(D^u_\varepsilon / \partial D^u_\varepsilon)$. Hence

$$\alpha = (B_{\mathfrak{b}} \circ c)^* (\beta)$$

And thereby, the generator only depends on $\mathfrak{b}$ concluding the proof. □

Example 3.1.19 (A Map of K-Groups)**.** Let $f : M \to \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$ and $p \in \mathrm{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \to q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0) is a pair for p. Assuming we have regular index pairs $(N_q.L_q)$ and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \to p) : K^{k-1}(N_p, L_p) \to K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{triple}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \mathrm{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \to \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q) \tag{10}$$

We can define this also with canonical isomorphism classes:

$$\underline{\Delta^k} : \bigoplus_{p \in \mathrm{Crit}_{k-1}(f)} K^{k-1}(\mathcal{IP}_p) \to \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(\mathcal{IP}_q)$$

define direkt sum of canonical isomorphism classes

Definition 3.1.20 (Regular Index Pairs and a Filtration on M)**.** As always let $f : M \to \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m. Let $\varphi_t : M \to M$ be the flow given by $-\mathrm{grad} f$. For $0 \le j \le k \le m$ define:

$$W(k, j) = \bigcup_{j \le \lambda_p \le \lambda_q \le k} W(q \to p).$$

There exists a filtration

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M, \tag{11}$$

such that (N_k, N_{j-1}) is a regular index pair for $W(k, j)$ for all $0 \le j \le k \le m$. If we define $\mathcal{IP}_i := \mathcal{IP}_{\mathrm{Crit}_i(f)}$ we can induce a filtration of conected simple systems:

$$\emptyset \to \mathcal{IP}_0 \to \mathcal{IP}_1 \to \cdots \to \mathcal{IP}_{m-1} \to \mathcal{IP}_m$$

Proof. The proof is done by Dietmar Salamon and can be checked in [Sal85] corollary 4.4. □

Remark 3.1.21. Now $\mathcal{IP}_k$ has a pointed spaces as a representative, that if of the form $\bigvee_{p\in\mathrm{Crit}_k}(N_p/L_p)$. Hence, $\tilde{K}^k(\mathcal{IP}_k)$ has a representative that omitts an isomorphism to $\bigoplus_{p\in\mathrm{Crit}_k(f)}\tilde{K}^k(N_p/L_p)$. Hence as canonical isomophism classes of rings we get the isomorphism:

$$\tilde{K}^k(\mathcal{IP}_k) \to \bigoplus_{q\in\mathrm{Crit}_k(f)} \tilde{K}^k(\mathcal{IP}_p)$$

and with this we can define the morphism $\underline{\Delta^k} : \tilde{K}^{k-1}(\mathcal{IP}_{k-1}) \to \tilde{K}^k(\mathcal{IP}_k)$. I think with this, i can give the definition of the following spectral sequence in the language of canonical isomorphism classes!

4 The Spectral Sequence using Exact Couples

Definition 4.0.1. Let M be a smooth manifold with Morse data and

$$\varnothing =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 3.1.20. Then we define the Groups

$$\begin{array}{lll} D^{p,q} & := K^{p+q}(N_p) & \text{and} \quad D := \bigoplus_{p,q} D^{p,q} \\ E^{p,q} & := K^{p+q}(N_p, N_{p-1}) & \text{and} \quad E := \bigoplus_{p,q} E^{p,q} \end{array}$$

Now we have the exact sequence from the pair $(N_p, N_p - 1)$:

$$\cdots \longrightarrow K^{p+q-1}(N_{p-1}) \xrightarrow{\delta^*_{p,q}} K^{p+q}(N_p, N_{p-1}) \xrightarrow{j^*_{p,q}} K^{p+q}(N_p) \xrightarrow{i^*_{p,q}} K^{p+q}(N_{p-1})\cdots$$

With this we define the maps

$$\begin{array}{lllc|l} \alpha & := \bigoplus_{p,q} & i^*_{p,q} : & D \to D & (-1,1) \\ \beta & := \bigoplus_{p,q} & \delta^*_{p,q} : & D \to E & (1,0) \\ \gamma & := \bigoplus_{p,q} & j^*_{p,q} : & E \to D & (0,0) \\ & & & & \text{bidegree} \end{array}$$

Obviously by definition the compositions in the triangle are exact:

$$\begin{tikzcd} D \arrow[rr, "\alpha"] & & D \arrow[dl, "\beta"] \\ & E \arrow[ul, "\gamma"] & \end{tikzcd}$$

Finally we define $d := \beta \circ \gamma$.

Definition 4.0.2 (Exact Couples)**.** Two groups D, E and maps

$$\begin{aligned} \alpha:&\ \ D \to D \\ \beta:&\ \ D \to E \\ \gamma:&\ \ E \to D \end{aligned} \qquad \begin{tikzcd} D \arrow[rr, "\alpha"] & & D \arrow[dl, "\beta"] \\ & E \arrow[ul, "\gamma"] & \end{tikzcd}$$

such that the compositions in the trianlge are exact are called an **exact couple**. we define $d := \beta \circ \gamma$. In this case it is obvious that $d^2 = 0$.

Definition 4.0.3 (Categorie of Exact Couples)**.** We define the categorie **ExCoup** as follows: The objects are exact couples, i.e. quintibles $(D, E, \alpha, \beta, \gamma)$ where D, E are groups (or objects in an abelian categorie), such that the triangle is exact:

$$\begin{tikzcd} D \arrow[rr, "\alpha"] & & D \arrow[dl, "\beta"] \\ & E \arrow[ul, "\gamma"] & \end{tikzcd}$$

.Given another exact couple $(D', E', \alpha', \beta', \gamma')$ we define a morphism from one to another to be a touple of morphisms $(f : D \to D', g : E \to E')$ such that the following diagram commutes:

$$\begin{tikzcd} D \arrow[r, "\alpha"] \arrow[d, "f"] & D \arrow[r, "\beta"] \arrow[d, "f"] & E \arrow[r, "\gamma"] \arrow[d, "g"] & D \arrow[d, "f"] \\ D' \arrow[r, "\alpha'"] & D' \arrow[r, "\beta'"] & E' \arrow[r, "\gamma'"] & D' \end{tikzcd}$$

Corollary 4.0.4 (An Endofunktor in the Categorie of Exact Couples)**.** We can derive a new exact couple with the following definitions:

$$\begin{aligned} E' &:= \ker d \big/ \mathrm{im} d \\ D' &:= \alpha(A) \end{aligned} \qquad \begin{aligned} \alpha' &: D' \to D' & a &\mapsto \alpha(a) \\ \beta' &: D' \to E' & a &= \alpha(\tilde{a}) \mapsto [\beta(\tilde{a})] \\ \gamma' &: E' \to D' & [b] &\mapsto \gamma(b) \end{aligned} \qquad \begin{tikzcd} D' \arrow[rr, "\alpha'"] & & D' \arrow[dl, "\beta'"] \\ & E' \arrow[ul, "\gamma'"] & \end{tikzcd}$$

We again defnie $d' := \beta' \circ \gamma'$. The derifing of a new exact couple as defined above defines an endofunctor which er denote by Der.

Proof. Let $(D, E, \alpha, \beta, \gamma)$ be the exact couple we started with and call the derived couple $(D', E', \alpha', \beta', \gamma')$. First we show the well definition of β. Let $a \in D' = \alpha(a_1) = \alpha(a_2)$. Then $a_1 - a_2 \in \ker \alpha = \mathrm{im}\gamma$. Hence $\beta(a_1) - \beta(a_2) = \beta(a_1 - a_2) \in \mathrm{im}(\beta \circ \gamma) = \mathrm{im}d$. Furthermore, since $\beta(a_1) \in \mathrm{im}\beta = \ker \gamma \subseteq \ker d$ we have that the map β' is well defined. For the well definition of γ' we ssume that $[b_1] = [b_2] \in E'$. But then the difference of b_1 and b_2 is zero in E' meaning in the image oif d, which is again contained in the image of β and thereby in the kernal of α. This menas that under α the image of b_1 and b_2 agree. Now to show that the derived couple is also exact one continuous the diagram yoga. We do exemplary the equality of the kernal of β' with the image of α': Assume that $\beta'(x) = 0$. That means that $\beta(\tilde{x}) \in \mathrm{im} d$ with $\alpha(\tilde{x}) = x$. We now want to show that $x \in \alpha^2 D$, and do this by showing that $\tilde{x} = \alpha(z)$ for some z. Now since $\beta(\tilde{x}) \in \mathrm{im} d$ we have that $\beta(\tilde{x}) = d(y) = \beta \circ \gamma(x)$ for some y. In other words, $\tilde{x} - \gamma(y) \in \ker \beta = \mathrm{im} \alpha$. Hence, $\tilde{x} - \gamma(y) = \alpha(z)$ for some z. But now applying α to both side and using that $\alpha \circ \gamma = 0$ we can deduce that $\alpha^2(z) = \alpha(\tilde{x})$. This shows that the kernal of β' is contained in the image of α'. The other direction of the inclusion is easier: Let $a = \alpha(\tilde{a}) \in D'$. Then $\beta' \circ \alpha'(z) = [\beta(z)] = \beta \circ \alpha(\tilde{a}) = 0$. Similary one checks for exactnes in the other corners. Now to show that we have an endofunktor we need to define what to do on morphisms. So assume that we have the two exact couples $(D, E, \alpha, \beta, \gamma)$ and $(\widetilde{D}, \widetilde{E}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$ with the morphisms $(f : D \to \widetilde{D}, g : E \to \widetilde{E})$. We now need to define a morphism of exact couples $((f' : D' \to \widetilde{D}', g : E' \to \widetilde{E}'))$ between the derived couples. For this we define:

$$f' : D' \to \widetilde{D}'; x \mapsto f(x) \quad \text{and} \quad g : E' \to \widetilde{E}'; [x] \mapsto [g(x)]$$

and check for well definition as maps and that the pair is a morphism of exact couples. The well definition of f' follows directly from $f \circ \alpha = \widetilde{\alpha} \circ f$. For the well definition of g' asume that $[x] = [y] \in E'$, meaning that $x - y \in \mathrm{im} d = \mathrm{im} \beta \circ \gamma$ hence for some z we have that $x - y = \beta \circ \gamma(z)$ and hence $g(x - z) = g \circ \beta \circ \gamma(z) = \widetilde{\beta} \circ f \circ \gamma(z) = \widetilde{\beta} \circ \widetilde{\gamma} \circ g(z) \in \mathrm{im} \widetilde{d}$. This proofes the well definition. □

Definition 4.0.5 (Spectral Sequence From an Exact Couple)**.** We can now iterate this procedure to get a list of spaces with maps $(E, d), (E', d'), (E'', d'') \ldots$ or often written as $(E_1, d_1), (E_2, d_2), \ldots$ which is called a **spectral sequence**. Notice how E_{r+1} is given by (E_r, d_r) whilst d_{r+1} needs further information. Whilst the spectral sequence is defined inductively, the needed information is not captured in $(E_r, d_r)_r$ but rather in the exact couple. Or said differently, we get a sequence of exact couples $(E_r, D_r, \alpha_r, \beta_r, \gamma_r)$ from which we can get the spectral sequence. We can reformulate the maps as follows:

$$D_r = \alpha^{r-1}(D), \quad E_r = \ker d_{r-1} / \mathrm{im} d_{r-1}, \quad \alpha_r = \alpha|_{\alpha^{r-1}(D)}, \quad \beta_r = [\beta \circ \alpha^{1-r}], \quad \gamma_r = \gamma, \quad d_r = \beta_r \circ \gamma_r$$

α^{1-r} means taking the preimage by alpha $1-r$ times. The well definition of this formulation was proven earlier. The bidegree of the maps is easy to deduct from the formulations above:

$$\begin{array}{lll} \alpha_r & \text{has bidegree} & (-1, 1) \\ \beta_r & \text{has bidegree} & (-r+2, r-1) \\ \gamma_r & \text{has bidegree} & (0, 0) \\ d_r & \text{has bidegree} & (r, +r+1) \end{array}$$

Corollary 4.0.6. We can define the categorie of spectral sequences **SpS** where the objects consist of families of pairs $(E_r, d_r)_r \in \mathbb{Z}$ where E_r are groups (or objects in an abelian categorie) and d_r are endomorphisms in E_r, such that $E_{r+1} \cong \frac{\ker d_r}{\mathrm{im} d_r}$. Morphisms between two spectral sequences $(E_r, d_r)_{r \in \mathbb{N}} \to (\widetilde{E_r}, \widetilde{d_r})_{r \in \mathbb{N}}$ are families of morphisms $(f_r : E_r \to \widetilde{E_r})$ such that for all r we have that $f_r \circ d_r = \widetilde{d_r} \circ f_r$.

Corollary 4.0.7. We have a functor from **ExCoup** to **SpS** given as follows: Let $(E, D, \alpha, \beta, \gamma)$ be an exact couple. Let $(E_r, d_r, \alpha_r, \beta_r, \gamma_r) = Der^r((E, D, \alpha, \beta, \gamma))$ where

Der^r denotes the r-th derivation of the exact couple. Then $(E_r, d_r := \beta_r \circ \gamma_r)_r$ is an exact sequence as described above. Assume that we have the two exact couples $(D, E, \alpha, \beta, \gamma)$ and $(\widetilde{D}, \widetilde{E}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$ with the morphisms $(f : D \to \widetilde{D}, g : E \to \widetilde{E})$. Then the famliy $(Der^r(f))_r$ is a morphism of spectral sequences.

Corollary 4.0.8 (Explicit Pages)**.** For a spectral seqquence arising from an exact couple defined as above we have a natural isomorphism

$$E_{r+1} = \frac{\gamma^{-1}(\alpha^r D)}{\beta(\ker \alpha^r)}$$

Proof. We proceed by induction starting with $n = 1$. Here we have by exactnes:

$$\ker(\beta \circ \gamma) = \gamma^{-1}(\ker(\beta)) = \gamma^{-1}(\mathrm{im}(\alpha)), \quad \text{and} \quad \mathrm{im}(\beta \circ \gamma) = \beta(\ker(\alpha)).$$

Hence:

$$E_2 = \frac{\ker(d_1)}{\mathrm{im}(d_1)} = \frac{\ker(\beta \circ \gamma)}{\mathrm{im}(\beta \circ \gamma)} = \frac{\gamma^{-1}(\alpha D)}{\beta(\ker(\alpha))}.$$

Now Assume that

$$E_r = \frac{\gamma^{-1}(\alpha^{r-1} D)}{\beta(\ker \alpha^{r-1})}$$

Then we have

$$\ker d_r = \ker \beta_r \gamma_r = \gamma_r^{-1}(\mathrm{im}(\alpha_r)) = \gamma_r^{-1}(\alpha^r(D)).$$

Now $\gamma_r : E_r \to D_r$ maps $[x] \mapsto \gamma(x)$, hence by induction assumption

$$\ker d_r = \frac{\gamma{-}1(\alpha^n D)}{\beta(\ker \alpha)}.$$

Notice that $\alpha^{r-1}(\ker(\alpha_r))$ containes those elements in D, that are killed by the α^n, in other words this equals $\ker \alpha^r$. Now $\beta_r = [\beta \circ \alpha^{r-1}]$ and hence using the induction assumptian again we can conclude:

$$\mathrm{im}(\beta_r \gamma_r) = [\beta(\alpha^{r-1} \ker(\alpha_r))] = \frac{\beta(\ker \alpha^r D)}{\beta(\ker \alpha^{r-1})}.$$

Now using the isomorphism theorem we get the natural isomorphism from the statement. □

Definition 4.0.9. In our case all pages of the spectral sequence inherit a double grading and the boundary maps to. Hence we will denote

$$E_r = \bigoplus_{p,q} E_r^{p,q} \quad \text{and} \quad d_r = \oplus_{p,q} d_r^{p,q}$$

Corollary 4.0.10. The grading carries through all pages and we can reformulate

$$E_{r+1}^{p,q} = \frac{\gamma^{-1}\alpha^r D}{\beta(\ker \alpha^r)} \cap E_{r+1}^{p,q} = \frac{\gamma^{-1}\big((\alpha^r D) \cap D^{p,q}\big)}{\beta((\ker \alpha^r) \cap D^{p-1,q})} = \frac{\gamma^{-1}\alpha^r D^{p+r,q-r}}{\beta(\ker \alpha^r|_{D^{p-1,q}})}$$

Corollary 4.0.11 (The Exact Sequence Stabilizes)**.** For all $r \geq m+1$ it holds that

$$E_r = E_{r+1} = \cdots$$

For this great enough r we define $E_\infty := E_r$.

Proof. Notice how our original $E^{p,q}$ is supportet in a strip regarding p which is fancy languare for saying that $E^{p,q} = 0$ for $p > 0$ and for $p \geq m$. These conditinos cary through the whole spectral sequence, as suceeding pages are quotients of preceeding ones respecting the grading. Now the map d_r has bidegree $(r, -r+1)$, and hence $d_r = 0$ since the domain or codomain is zero , if the p grading is greater then the p–length of the supporting strip meaning $p > m$ which gives us the claimes $r \geq m+1$. □

Now we refomulate the corollary 4.0.8 for the infinity page. We have the constant sequence of groups $E_r = E_{r+1} = \cdots$ and by the corollary we have

$$\frac{\gamma{-1}(\alpha^r D)}{\beta(\ker \alpha^r)} = \frac{\gamma{-1}(\alpha^{r+1} D)}{\beta(\ker \alpha^{r+1})} = \frac{\gamma{-1}(\alpha^{r+2} D)}{\beta(\ker \alpha^{r+2})} = \cdots$$

Hence in our later inspections of E_∞ we can assume r to be arbitrary big.

Lemma 4.0.12 (The last page)**.** *In our spectral sequence, we have that*

$$E_\infty^{p,q} = \frac{\ker[K^{p+q}(M) \to K^{p+q}(N_{p-1})]}{\ker[K^{p+q}(M) \to K^{p+q}(N_{p1})]}$$

Proof. For each fixed (p,q) there is an r big enough such that $\alpha^r D^{p,q} = 0$, since α^r has degree $-r$ in p, hence for $r > p$, α^r maps $D^{p,q}$ to something negative in p and hence to the trivial group. Thus $D = \cup_{r\geq 0} \ker \alpha^r$. Now the denominator of $E_r^{p,q}$ equals $\beta(\ker(\alpha^r|_{D^{p+q-1,q-r}})$ and hence for r big enough this equals the image of $\beta|_{D^{p-1,q}}$ which in turn is the kernal of $\gamma|_{E^{p,q}}$. For the numerator, we start by looking at $(\alpha^r D)^{p,q} = \alpha^r D^{p+r,q-r}$. Now $D^{p+r,q-r} = K^{p+q}(N_{p+n})$. Hence for n big enough (such that $N_{p+n} = M$) we can calculate $\alpha^n(K^{p+q}(N_{p+n})) = \operatorname{im}(K^{p+q}(M) \to K^{p+q}(N_p))$. This results in the description

$$E_\infty^{p,q} = \frac{\gamma^{-1}\big(\operatorname{im}(K^{p+q}(M) \to K^{p+q}(N_p))\big)}{\ker \gamma|_{E^{p,q}}}$$

is das wahr mit gamma eingeschränkt auf Epq?

Next we define the groups

$$L^{p,q} := \operatorname{im}(K^{p+q}(M) \to K^{p+q}(N_p)) \cap \operatorname{im}\gamma = \operatorname{im}(K^{p+q}(M) \to K^{p+q}(N_p)) \cap \ker \alpha$$

And since we have for any f and any subset S of the codomain of f the equality $f^{-1}(S) = f^{-1}(S \cap \operatorname{im} f)$ holds, we have in our setting $E_\infty^{p,q} = \frac{L^{p,q}}{\ker(\gamma|_{E^{p,q}})}$. Now the kernal of the funktion

$$\tilde{\gamma} : \gamma^{-1}(L^{p,q}) \to L^{p,q};\ x \mapsto \gamma(x)$$

equals $\ker \gamma|_{E^{p,q}}$: Since $\gamma^{-1}(L^{p,q}) \subseteq E^{p,q}$ the kernal of $\tilde{\gamma}$ is clearly contained in $ker\ \gamma|_{E^{p,q}}$. If on the other hand $\gamma(x) = 0$ for $x \in E^{p,q}$ then $x \in \gamma^{-1}(0)$ and since zero is contained in $L^{p,q}$ we conclude that the two kernals are equal and hence we can apply the isomorphism theorem to get the isomorphism that is induced from γ:

$$E_\infty^{p,q} = \frac{L^{p,q}}{\ker(\gamma|_{E^{p,q}})} \to L^{p,q}\,.$$

Now finally we define the mapping

$$\ker\left(K^{p+q}(M) \to K^{p+q}(N_{p-1})\right) \to L^{p,q} = \operatorname{im}(K^{p+q}(M) \to K^{p+q}(N_p)) \cap \ker \alpha$$

That maps an element $x \in K^{p,q}(M)$ to the pullback of the inclusion $N_p \to M$. This map is well defined, since applying α to the image of an element from the domain is the same as the pullback of x via the inclusion $N_{p-1} \to M$ and equlals zero by definition of the domain. Hence the original map maps into the kernal of alpha which proofs the well definition. This line of reasoning also shows the surjectivity since every element in $K^{p,q}(N_p)$ that comes from the whole M and maps to zero if restricted to N_{p-1} can be expressed as the pullback from the whole M restricted to those, that vanish over N_{p-1}. TIts easy to see that the kernal of this surjection equals the kernal of the pullback

$$K^{p,q}(M) \to K^{p,q}(N_p)\,,$$

because ever element that pulls back to zero over N_p also pulls back to zero over N_{p-1}, and the other containemend is trivially true. To the surjection we can again apply the isomorphism theorem to get the isomorphism

$$\frac{\ker\left(K^{p+q}(M) \to K^{p+q}(N_{p-1})\right)}{\ker\left(K^{p,q}(M) \to K^{p,q}(N_p)\right)} \cong L^{p,q}\,,$$

that is induced by the pullback of the inclusion. Together with the γ induced isomorphism from $E_\infty^{p,q} \to L^{p,q}$ we conclude the proof. □

Definition 4.0.13 (Associated Graded Group)**.** Let G be a group together with a finite filtration of subgroups $\{F^i\}_{i=-1,\dots,-m}$ for $i = -1, \dots m$ such that

$$G = F^{-1} \subseteq F^0 \subseteq F^1 \subseteq \cdots \subseteq F^{m-1} \subseteq F^m = 0$$

Then we define the **associated graded group of G with respect to the Filtration $\{F_i\}$** to be

$$\operatorname{gr}(G) := \bigoplus_{i=0}^{m} F^{i-1}/F^i\,.$$

From $\mathrm{gr}(G)$ we can rebuild G by rebuilding $\{F^i\}$ up to extension problems in the following way:

$$\begin{array}{lll}
F^{m-1} & = F^{m-1}/F^m = \mathrm{gr}(G)_m & \\
F^{m-2} & \text{is an extension} & 0 \longrightarrow F^{m-1} \longrightarrow F^{m-2} \longrightarrow F^{m-2}/F^{m-1} = \mathrm{gr}(G)_{m-1} \longrightarrow 0 \\
F^{m-1} & \text{is an extension} & 0 \longrightarrow F^{m-2} \longrightarrow F^{m-3} \longrightarrow F^{m-3}/F^{m-2} = \mathrm{gr}(G)_{m-2} \longrightarrow 0 \\
\vdots & \cdots & \\
F^{-1} & \text{is an extension} & 0 \longrightarrow F^{0} \longrightarrow F^{-1} \longrightarrow F^{-1}/F^{0} = \mathrm{gr}(G)_{0} \longrightarrow 0
\end{array}$$

Corollary 4.0.14. Define the groups

$$F^l K^n(M) := \ker\left(K^n(M) \to K^n(N_l)\right) .$$

Those give an decending filtration of the $n-th$ K-group of M:

$$K^n(M) = F^{-1}K^n(M) \supseteq F^0K^n(M) \supseteq \cdots \supseteq F^{m-1}K^n(M) \supseteq F^mK^n(M) = 0 .$$

Then by the corollary above we can conculde that

$$E_r^{p,q} = \frac{F^{p-1}K^{p+q}(M)}{F^pK^{p+q}(M)} .$$

Hence, the diagonals on the infinity page are the associated graded groups to $K^{p+q}(M)$ with respect to the filtration above.

Corollary 4.0.15 (The first page)**.** In our setting we get the first page to be defined as the quotient

$$E^{p,q} := K^{p+q}(N_p, N_{p-1})$$

Now since (N_p, N_{p-1}) is a regular index pair for $\mathrm{Crit}_p(f)$ we have a flow induced homotopiy equivalentce:

$$N_p/N_{p-1} \simeq \Big(\bigcup_{w\in Crit_p(f)} D^u_\varepsilon(w)\Big)/\Big(\bigcup_{w\in Crit_p(f)} \partial D^u_\varepsilon(w)\Big) \cong \bigvee_{w\in \mathrm{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D^u_\varepsilon(w) \;:=\; \varphi_w^{-1}\big(\{v \in T^u_wM \mid \|v\| \le \varepsilon\}\big),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$\begin{aligned}
E_1^{p,q} &\cong \widetilde{K}^{p+q}(N_p/N_{p-1}) && (12)\\
&\cong \bigoplus_{w\in \mathrm{Crit}_p(f)} \widetilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \widetilde{K}^q(pt)) && (13)\\
& && (14)
\end{aligned}$$

Notice that this is trivial for all odd q and naturally isomorphic for all even q. In sum we have the chain of horizontal isomorphisms:

$$\begin{tikzcd}
E_1^{p,q} \arrow[r, "\alpha"] \arrow[d, "d_1^{p,q}"] & \widetilde{K}^{p+q}(N_p/N_{p-1}) \arrow[r, "\beta"] \arrow[d] & \bigoplus_{w\in\mathrm{Crit}_p(f)} \widetilde{K}^q(pt) \arrow[d] & C^p(M,\mathfrak{A},\widetilde{K}^q(pt)) \arrow[l] \arrow[d] \\
E_1^{p+1,q} \arrow[r, "\alpha'"] & \widetilde{K}^{p+1+q}(N_{p+1}/N_p) \arrow[r, "\beta'"] & \bigoplus_{w\in\mathrm{Crit}_{p+1}(f)} \widetilde{K}^q(pt) & C^{p+1}(M,\mathfrak{A},\widetilde{K}^q(pt)) \arrow[l]
\end{tikzcd}$$

Diagram 15

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology? We already answered this in theorem **??**. This theorem lets us define a new spectral sequence as follows:

Definition 4.0.16. Let $(D, E, \alpha, \beta, \gamma)$ be the exact couple from definition 4.0.1. Furthermore, let $\Psi := \oplus_{p,q}\Psi^{p,q}$ be the isomorphism from $E = \bigoplus_{p,q} E^{p,q} \to \bigoplus_{p,q} C^p(M, \mathfrak{A}, \widetilde{K}^q(pt))$ that is defined in equation 12. Furthermore, let $\beta : K^{2s}(pt) \to \mathbb{Z})$ be the isomorphisms induced from the bott generator. Then we define

$$\Lambda : E \to \widetilde{E} := \bigoplus_{p,q} C(M, \mathfrak{A}, \mathbb{Z}) \otimes_{\mathbb{Z}} \begin{cases} \mathbb{Z} & \text{if } q \text{ is even,} \\ 0 & \text{if } q \text{ is odd.} \end{cases}$$

Then define $\widetilde{D} := D$ and the new exact couple $(\widetilde{D}, \widetilde{E}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$ as follows:

$$\begin{array}{lll} \widetilde{\alpha} & := \alpha & : \ \widetilde{D} \to \widetilde{D} \\ \widetilde{\beta} & := \Lambda \circ \beta & : \ \widetilde{D} \to \widetilde{E} \\ \widetilde{\gamma} & := \gamma \circ \Lambda^{-1} & : \ \widetilde{E} \to \widetilde{D} \end{array}$$

We get a morphism of exact couples from $(D, E, \alpha, \beta, \gamma)$ to $(\widetilde{D}, \widetilde{E}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$ that is given as (Λ, id_D). This new exact couple gives rise to a spectral sequence $(\widetilde{E}_r, \widetilde{d}_r)_r$, where the first page is the Morse-chain group in all even q. Hence, the second page is the Morse homology.

Corollary 4.0.17 (Periodicity in q)**.** The exact couple $(\widetilde{D}, \widetilde{E}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$ has periodicity in q meaning that we have the isomophism of exact couples given by the identity functions

$$f^{p,q} : \widetilde{E}p, q \to \widetilde{E}^{p,q+2}$$

Here the notation might seem a bit strange. However, the summonds $\widetilde{E}p, q$ and $\widetilde{E}p, q + 2$ are actually the same. They only differ by there inclsion map into $\widetilde{E}$. Now the pair $(f : \oplus_{p,q} : \widetilde{E} \to \widetilde{E}, g := \mathrm{id}\,\widetilde{D} \to \widetilde{D})$ defines an isomorphism of exact couples. The compatibility with morphisms is clear. Now since building up the spectral sequence is functorial we know that for all p, q we have canonical (in the sence that they traslate $\widetilde{d}_r^{p,q}$ to $\widetilde{d}_r^{p,q+2}$) isomorphisms

$$\widetilde{E}_r^{p,q} \cong \widetilde{E}_r^{p,q+2}.$$

Furthermore, $\widetilde{E}_r^{p,q} = 0$ for all odd q.

Definition 4.0.18. For $r > 0$ we define $EM_r^p := \widetilde{E}_r^{p,0}$ and

$$dM_r : EM_r^p \to EM_r^{p+r}; x \mapsto \begin{cases} 0 & \text{if } r \text{ is even} \\ \widetilde{d_r^{p,0}}(x) \in \widetilde{E}_r^{p,-r+1} = EM_r^p & \text{else.} \end{cases}$$

The last equality comes from the corollary above. Furthermore, we have a morphism of spectral sequences $(\widetilde{E_r}, \widetilde{d_r}) \to (EM_r, dM_r)$ by the maps

$$f_r^{p,q} : \widetilde{E}^{p,q} \to EM_r^p; x \mapsto \begin{cases} 0 & \text{if } q \text{ is odd} \\ x \in EM_r = \widetilde{E}^{p,0} \cong \widetilde{E}^{p,q} & \text{if } q \text{ is even.} \end{cases}$$

Theorem 4.0.19 (Summary of the Specrtal Sequence)**.** *For each Morse datum $(M, f, g, \mathfrak{A})$ together with a filtration $N_{-1} \subseteq N_0 \subseteq \cdots \subseteq N_{m-1} \subseteq N_m = M$ from (11) there esists a Spectralsequence (EM_r^p, dM_r^p) where each page is graded in p such that*

- *The first page is the morse complex, hence the second page is the Morse homology*
- *The spectral sequence becomes stationary and the infinity page is the associeted graded goup of the K-groups meaning:*

$$\bigoplus_{p \in 2\mathbb{N}} E_\infty^p \text{ is a associated graded group to } K^0(M)$$
$$\bigoplus_{p \in 2\mathbb{N}+1} E_\infty^p \text{ is a associated graded group to } K^1(M).$$

Remark 4.0.20 (Recover from Associated Graded Groups)**.** Let G be a group and

5 Higher Differentials

6 Examples

We begoin with a minimal example that has non trivial higher differential:

Example 6.0.1 (The Sphere)**.** If we take the three-sphere $S^3 \subseteq \mathbb{R}^4$ and denote the north pole with N and the south pole with S we get the Morse chain complex as

$$0 \longrightarrow \langle S\rangle_{\mathbb{Z}} \longrightarrow 0 \longrightarrow 0 \longrightarrow \langle N\rangle_{\mathbb{Z}} \longrightarrow 0$$

Then for the second page we get the Morse homology together with the 0 maps and on the third page we have:

$$\begin{tikzcd} \langle S\rangle & 0 & 0 & \langle N\rangle \arrow[lll, bend left, "d_3^0"] \end{tikzcd}$$

Now we want to check what $d_3^0(S)$ is. For this we want to use its definition and hence we need to give the Morse filtration. For this we can use:

$$N_{-1} \coloneqq \varnothing, \quad N_{0,1,2} \coloneqq H_{-} \text{ the southern hemisphere}, \quad N_3 = S^3 \,.$$

First we translate E_r into the language where the definition was formulated. We know that EM_3^p is naturally isomorphic to

$$E_3^{p,0} = \frac{\gamma^{-1}\big(\alpha^2 D^{p,-2}\big)}{\beta\big(\ker(\,\alpha^2|_{D^{p+3-1,-2}})\big)}$$

Now for $r = 3$ and $p = 0$

$$\alpha^2 D^{2,-2} = \operatorname{im}(K^0(N_2) \to K^0(N_0)) = \operatorname{im}(K^0(H_{-}) \to K^0(H_{-})) \cong Z$$

Hence $E_3^{0,0} = \gamma^{-1}(\alpha^2 D^{2,-2})^{0,0} \cap E_0^{0,0} = K^0(N_0, \varnothing) \cong \mathbb{Z}$. And similar we conclude that $E_3^{3,0} = K^3(N_3, N_2) \cong \mathbb{Z}$. Now we analyze the map

$$d_3^{3,0} : K^3(N_3, N_2) \to K^0(N_0, \varnothing)$$

So far we where able to ignore all orientation, (which we formally need for the morse boundary operator). So lets fix the orientations of the unstable part by saying the unstable part of the north pole (i.e. the whole S^3) is the canonical orientation of $S^3 \supset \mathbb{R}^4$, whilst the unstable part of the south pole (i.e. the south pole) has the orientation -1.

Now $d_r = \beta \circ \alpha^2 \circ \gamma$ and $\gamma : K^3(N_3, N_2) \to K^0(N_3)$ maps our generator induced by N to the standard generator of $K^3(S^3)$ induced by the choosen orientation, due to N_2 being a contractible subset. Now $\alpha^2 : K^3(N_3) \to K^3(N_1)$ is the zero map, since its range is 0 due to N_1 being homotopically equivalent to the point and 3 being odd.Finally $\beta : K^3(N_1) \to K^4(N_0, \varnothing)$ is the connecting homomorphism that trivial here. Hence $d^3, 0_3$ is the zero map.And thereby EM_4 which equalse E_∞ is

$$\langle S\rangle \qquad 0 \qquad 0 \qquad \langle N\rangle$$

Now how to we get information about our beloved K groups back? For this we remined ourself about the filtration

$$K^0(S^3) = F{-1}K^0(S^3) \supseteq F^0K^0(S^3) \supseteq F^1K^0(S^3) \supseteq F^2K^0(S^3) \supseteq F^3K^0(S^3) = 0$$

Now using this filtration we have the short exact sequence

$$0 \longrightarrow F^0K^0(S^3) \longrightarrow K^0(S^3) \longrightarrow E_4^{0,0} = \tfrac{F^{-1}K^0(S^3)}{F^0K^0(S^3)} \longrightarrow 0$$

Hence, $K^0(S^3)$ is an extension of $F^0K^0(S^3)$ and $E_4^{0,0} \cong \mathbb{Z}$ Furthermore, we have the exact sequence

$$0 \longrightarrow F^1K^0(S^3) \longrightarrow F^0K^0(S^3) \longrightarrow E_4^{1,-1} = \tfrac{F^0K^0(S^3)}{F^1K^0(S^3)} = 0 \longrightarrow 0$$

Hence, $F^0K^0(S^3) \cong F^1K^0(S^3)$ and by the same argument we can continue this chain

$$F^0K^0(S^3) \cong F^1K^0(S^3) \cong F^2K^0(S^3) \cong F^3K^0(S^3) = 0$$

Hence, the first short exact sequence gives is an isomorphism between $K^0(S^3) \cong \mathbb{Z}$.

Now we want to calculate the K group of a non trivial example:

Example 6.0.2 (The Real Projective Space)**.** Here, we define $\mathbb{RP}^n$ to be $\mathbb{R}^{n+1}\setminus 0$ modulo scalar multiples. Hence it is the space of lines in $\mathbb{R}^{n+1}$. Alternatively we can view it as the n-sphere where the operation from $\{\mathrm{id}, -\mathrm{id}\}$ is quotient out. From this description we get a metric, since $S^n \supseteq \mathbb{R}^{n+1}$ inherits the subspacae metric in which all operations from $\{\mathrm{id}, -\mathrm{id}\}$ are isometries. Hence, this metric projects down to $\mathbb{RP}^n$. Now let

$$A = \mathrm{diag}(\lambda_0, \lambda_1, \dots, \lambda_n)$$

be a symmetric matrix with real $\lambda_i \in \mathbb{R}$. Then we define the map

$$\Lambda : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}; \Lambda(x) := \frac{\langle x, Ax\rangle}{\|x\|^2}\,.$$

Notice that this function descends to a well defined function $\tilde{\Lambda} : \mathbb{RP}^n \to \mathbb{R}$. On the whole $\mathbb{R}^{n+1}$ we can define the function

$$f : \mathbb{R}^{n+1} \to \mathbb{R};\ f(x) := \frac{1}{2}\langle x, Ax\rangle = \frac{1}{2}\sum_{i=0}^{n} \lambda_i x_i^2\,.$$

Now assume that $\varphi_t(x)$ is the flow associated to $-\mathrm{grad}(f)$, meaning

$$\frac{\mathrm{d}\varphi_t(x)}{\mathrm{d}t} = -\mathrm{grad}_{\varphi_t(x)}(f) = -A\varphi_t(x)\,,$$

since

$$\frac{\partial f}{\partial x_i}(x) = \lambda_i x_i\,.$$

Now we have the lemma:

Lemma 6.0.3. *The flow lines of the negative gradient flow with respect to $\frac{1}{2}\tilde{\Lambda}$ on $\mathbb{RP}^n$ are the images under the quotient map*

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{RP}^n$$

of the flow lines of the linear system

$$\frac{\mathrm{d}\Phi_t(x)}{\mathrm{d}t} = -A\Phi_t(x)$$

Proof. Let x be a point on the flow line $\Phi_t(x)$. Then $-\mathrm{grad}(f)(x) = -Ax$. Now we split this into a part that is orthogonal to y and one that is paralell:

$$-\mathrm{grad}(f)(x) = \underbrace{-\Lambda(x)x}_{\text{parallel to } x} + \underbrace{-Ax + \Lambda(x)x}_{\text{orthogonal to } x}.$$

The claim of orthogonality is just a calculation of the inner product $\langle -Ax + \Lambda(x)x, x\rangle = 0$ □

Now we radially project this onto the sphere via

$$\pi_r : \mathbb{R}^{n+1} \setminus \{0\} \to S^n; x \mapsto \frac{x}{\|x\|}$$

the differential of this projection kills the parallel part and hence reads:

$$\mathrm{d}\pi_r|_x \left(-\mathrm{grad}_x(f)\right) = \frac{-Ax + \Lambda(x)x}{\|x\|}$$

If we restrict $f|_{S^n}$ we get that the gradient $-\mathrm{grad}_y(f|_{S^n}) = -Ay + \Lambda(y)y$. To see this, let $v \in T_yS^n \subseteq T_yR^{n+1}$ and remember that the inner produkt on S^b is induced from the euclidean one. Then

$$\begin{aligned}\langle \mathrm{grad}_y(f|_{S^n}), y\rangle_{S^n} &= \mathrm{d}\, f|_{S^n}(v) = \mathrm{d}f(v) = \langle \mathrm{grad}_y(f), v\rangle_{\mathbb{R}^{n+1}} \\ &= \langle \mathrm{proj}_{T_yS^n}(\mathrm{grad}_y(f)), v\rangle_{\mathbb{R}^{n+1}} = \langle Ay - \Lambda(y)y, v\rangle_{\mathbb{R}^{n+1}}\end{aligned}$$

Here $\mathrm{proj}_{T_yS^n}$ denotes the projection onto the subspace $T_yS^n \subseteq T_y\mathbb{R}^{n+1}$. This however tells us that the image $\widetilde{\Phi_t(x)}$ of $\Phi_t(x)$ under the radial projection solves the equation

$$\frac{\mathrm{d}\widetilde{\Phi_t(y)}}{t} = -\mathrm{grad}_y(f|_{S^n})(\widetilde{\Phi_t(y)}) = -\frac{1}{2}\mathrm{grad}_y(\Lambda|_{S^n})(\widetilde{\Phi_t(y)}) \quad \text{since } f = \frac{1}{2}\Lambda$$

Hence descending to the quotient $\mathbb{RP}^n$ yields that $\left[\widetilde{\Phi_{2t}(y)}\right]$ is a solution of the flow associeted to the negative gradient of $\tilde{\Lambda}$ in $\mathbb{RP}^n$. Now the other way round should be trivial: Take a solution $\psi_t(x)$ of $\tilde{\Lambda}$ in $\mathbb{RP}^n$. Then for fixed t, x let U be a neighbourhood of $\psi_t(x)$. This U can be lifted to S^n and the solution there solves the lifted version of the flow induced from $\tilde{\Lambda}$. And from here we can argue that the solution is part of a solution of the linear differential equation

$$\frac{\mathrm{d}\varphi_t(x)}{\mathrm{d}t} = -\mathrm{grad}_{\varphi_t(x)}(f) = -Ax\,.$$

This concludes the proof. Hence critical points in $\mathbb{RP}^n$ are eigenvalues of A. Hence, for the function to be Morse, we need the critical points to be isolated, meaning that all eigenspaces have dimension one. So we assume that the diagonal entries in A are all different. For simplicity we assume that $\lambda_0 < \lambda_1 \cdots < \lambda_n$. Since A is diagonal we know that $[e_i]$ are the critical points.

Lemma 6.0.4. *Each critical point $[e_i]$ is non degenerate, of index i and its unstable manifold is*

$$W([e_i] \to) =: U_i = \{[x_0 : x_1, \dots x_i, 0, \dots, 0] \in \mathbb{RP}^n | x_i \neq 0\}\,,$$

whilst the stable manifold is

$$W(\to [e_i]) =: S_i = \{[0, \dots, 0, x_i, x_{i+1}, \dots, x_n] \in \mathbb{RP}^n | x_i \neq 0\}$$

Proof. Let B^n be the open unit ball in $\mathbb{R}^n$ and $V_i := \{[x_, \dots, x_n] \in \mathbb{RP}^n | x_i \neq 0\}$. Then we define the parametrization

$$\begin{aligned} \varphi_i : B^n &\to V_i \\ (y_1, \dots y_n) &\mapsto [y_1, \dots, y_i, \sqrt{1 - \|y\|_2^2}, y_{i+1}, \dots, y_n]\,. \end{aligned}$$

Now the image gives representatives of norm one and hence in this chart we calculate:

$$\begin{aligned} \tilde{\Lambda} \circ \varphi_i(y_1, \dots, y_n) &= \sum_{j=1}^{i} y_i^2 \lambda_{j-1} + \lambda_i (1 - \|y\|_2^2) + \sum_{j=i+1}^{n} y_j^2 \lambda_j \\ &= \sum_{j=1}^{i} y_i^2 \lambda_{j-1} + \lambda_i (1 - \sum_{j=1}^{n} y_j^2) + \sum_{j=i+1}^{n} y_j^2 \lambda_j \\ &= \sum_{j=1}^{i} y_i^2 (\lambda_{j-1} - \lambda_i) + \lambda_i + \sum_{j=i+1}^{n} y_j^2 (\lambda_j - \lambda_i) \end{aligned}$$

With this local description we can calculate the hession in $[e_i] = \varphi_i(0)$:

$$\left.\frac{\partial^2 \tilde{\Lambda} \circ \varphi_i}{\partial y_k \, \partial y_l}\right|_0 = \begin{cases} 0 & \text{if } k \neq l \\ 2(\lambda_{k-1} - \lambda_i) < 0 & \text{if } k = l \leq i \\ 2(\lambda_k - \lambda_i) > 0 & \text{if } k = l > i \end{cases}$$

Hence, $[e_j]$ is a non degenerate critical point of index i. Now concerning the unstable manifolds, we pick a point $x = x_k e_k + \cdots x_l e_l \in \mathbb{R}^{n+1}$ with x_k and x_l not equal to zero. This points trajectory in $\mathbb{R}^{n+1}$ under $-\mathrm{grad}(f)$ is

$$\varphi_t(x) = x_k \mathrm{e}^{-\lambda_k t} e_k + \cdots x_l \mathrm{e}^{-\lambda_l t} e_l \,.$$

Hence for t approaching negative infinity, the term $x_l \mathrm{e}^{-\lambda_l t} e_l$ is leading, whilst for t approaching infinity, the summand $x_k \mathrm{e}^{-\lambda_k t} e_k$ is leading. Since projectively we don't care about absolute values but rater relations having a leading coordinate meaning that relatively all other coordinates approach zero. Hence:

$$\lim_{t\to-\infty}([\varphi_t(x)]) = [e_l] \quad \text{and,} \quad \lim_{t\to+\infty}([\varphi_t(x)]) = [e_k]\,.$$

$\square$

Definition 6.0.5 (Orientations)**.** For orienting the unstable manifolds we make use of the projection $\mathbb{R}^{n+1} \supseteq S^n \to \mathbb{RP}^n$. Hence, we can lift $T_{[e_i]}\mathbb{RP}^n$ to $T_{\pm e_i}S^n$ and include that into $T_{\pm e_i}\mathbb{R}^{n+1}$, where we have the maps $Q_{\pm e_i} : \mathbb{R}^{n+1} \to T_{\pm e_i}\mathbb{R}^{n+1}$. In this way we get two canonical lifts of the tangent space:

$$\begin{tikzcd} T_{[e_i]}\mathbb{RP}^n \arrow[r, "\mathrm{d}\pi_{e_i}^{-1}"] \arrow[d, "\mathrm{d}\pi_{-e_i}^{-1}"] & T_{e_i}S^n \arrow[d, "Q_{e_i}^{-1}"] \\ T_{-e_i}S^n \arrow[r, "Q_{-e_i}"] & \langle (e_j)_{j\neq i} \rangle \end{tikzcd}$$

In general, the diagram is not commutative. To see this notice how $\pi \circ -\mathrm{id} = \pi$. Hence $\mathrm{d}\pi|_{-e_i} \circ \mathrm{d} - \mathrm{id}|_{e_i} = \mathrm{d}\pi|_{e_i}$ and with that:

$$\begin{aligned} Q_{-e_i}^{-1} \circ \mathrm{d}\pi|_{-e_i}^{-1} \circ \mathrm{d}\pi|_{e_i} \circ Q_{e_i} &= Q_{-e_i}^{-1} \circ \mathrm{d}\pi|_{-e_i}^{-1} \circ \mathrm{d}\pi|_{-e_i} \circ \mathrm{d}(-\mathrm{id})|_{e_i} \circ Q_{e_i} \\ & Q_{-e_i}^{-1} \circ \mathrm{d}\pi|_{-e_i}^{-1} \circ \mathrm{d}\pi|_{-e_i} \circ Q_{-e_i} \circ -\mathrm{id} \\ &= Q_{-e_i}^{-1} \circ Q_{-e_i} - \mathrm{id} = -\mathrm{id}\,. \end{aligned}$$

Hence, we can define a canonical orientation on $T_{[e_i]}\mathbb{RP}^n$ that is given by $(e_j)_{j\neq i}$ or equivalent $(-e_j)_{j\neq i}$ depending on the choosen lift. Now in the same way we can induce orientations to the unstable manifolds U_i by two equivalent ways:

$$\begin{tikzcd} T_{[e_i]}U_i \arrow[r, hook] & T_{[e_i]}\mathbb{RP}^n \\ \langle (e_0, \dots e_{i-1}) \rangle \arrow[r, hook] \arrow[u] & T_{e_i}S^n \arrow[u] \end{tikzcd} \quad \text{or equivalently} \quad \begin{tikzcd} T_{[e_i]}U_i \arrow[r, hook] & T_{[e_i]}\mathbb{RP}^n \\ \langle -e_0, \cdots - e_{i-1} \rangle \arrow[r, hook] \arrow[u] & T_{-e_i}S^n \arrow[u] \end{tikzcd}$$

Now we know how the groups in the Morse chain complex are looking. The next step is the boundary operator:

Lemma 6.0.6. *The differential in $C^*(\mathbb{RP}^n, \Lambda^*)$ with the right orientations act as*

$$\partial_i([e_i]) = n([e_{i-1}], e_i)[e_{i+1}] = ((-1)^i - 1)[e_{i+1}]$$

Proof. First we notice that $W([e_i] \to [e_{i-1}])$ is a one dimensional projective space with two points missing. If we would look at the Morse homology with coeficients in $\mathbb{Z}_2$ we would be done here, since $\mathbb{RP}^1 \setminus \{p, q\}$ consists of two lines. Hence, over $\mathbb{Z}_2$ the boundary operator is trivial. However we want to use the orientations from above. Now let

$$x \in W([x_i] \to [e_{i-1}]) = \{[0 : \cdots 0 : x_{i-1} : x_i : 0 : \cdots 0] | x_{i-1} \neq 0 \text{ and } x_i \neq 0\}.$$

Then we can choose representatives, such that $x_{i-1}^2 + x_i^2 = 1$, and (x_{i-1}, x_i) is in the upper halve plane meaning that $x_i > 0$. Now take a $x \in W([e_i] \to [e_{i-1}])$. Then $T_x W([e_i] \to [e_{i-1}])$ has an oriented basis given as the pushforeward of $v = x_i e_{i-1} - x_{i-1} e_i$ under the projection $T_x S^n \to T_x \mathbb{RP}^n$. Now we want to analyze how this relates to the orientation induced by the flow and the two unstable orientations: if both x_i and x_{i-1} are positive, then v points in the direction of the flow or said different, $-\mathrm{grad}_x(f) = v$. Else, if x_{i-1} is negative, $-v$ points in the direction of the flow(i.e. $-\mathrm{grad}_x(f) = -v$). The parallel transport of $T_{[e_{i-1}]} U_{i-1}$ to the normal space of $-\mathrm{grad}_x(\tilde{\Lambda})$ induced by the negative flow maps the basis $(e_0, \ldots, e_{i-2})$ of $T_{[e_{i-1}]} U_i$ to $(e_0, \ldots e_{i-2})$ in the orthogonal complement of $-\mathrm{grad}_x(\tilde{\Lambda})$ in $T_x \mathbb{RP}^n$. So we need to compare in $T_x \mathbb{RP}^n$ the basis $(-\mathrm{grad}_x(f), e_0, \ldots e_{i-2})$ to the basis $(e_0, \ldots e_{i-1})$ in $T_{[e_i]} U_i$ after transporting it to $T_x \mathbb{RP}^n$ along the flow. This latter transport maps the basis vectors as follows:

$$e_j \mapsto \begin{cases} x_i e_{i-1} - x_{i-1} e_i & \text{if } j = i - 1 \\ e_j & \text{else.} \end{cases}$$

In other words, we need to ask when the two basis

$$((x_i e_{i-1} - x_{i-1} e_i), e_0, \ldots e_{i-2}) \quad \text{and} \quad (e_0, \ldots e_{i-2}, x_i e_{i-1} - x_{i-1} e_i),$$

if $x_{i-1} > 0$ are comparable.Then, the basis carry the same orientation, if and only if i is odd, since then it takes evenly many transpositions of coordinates.

If $x_{i-1} < 0$ we notice, that our representation of $[e_{i-1}]$ is given by $-e_{i-1}$ and hence the unstable tangent space has the basis $-e_0, \cdots - e_{i-2}$. Hence in x we get the oriented basis

$$(-\mathrm{grad}_x(f), -e_0, \cdots - e_{i-2}) = (-v, -e_0, \cdots - e_{i-2}).$$

We compare this basis to basis that comes from transporting the basis $(e_0, \ldots e_{i-1})$ of $T_{[e_i]} U_{[e_i]}$ to x resulting in the basis $(e_0, \ldots e_{i-2}, v)$. Hence we compare the bases

$$(-v, -e_0, \cdots - e_{i-2}) \quad \text{and} \quad (e_0, \ldots e_{i-2}, v)$$

we manipulate the first and keep note of the sign of the alterations:

$$(-v, -e_0, \cdots - e_{i-2}) = (-1)^i (v, e_0, \cdots e_{i-2}) = (-1)^{i+i-1} (e_0, \ldots e_{i-2}, v).$$

Hence, the orientations never agree. Now if i is odd the two flow lines cancel out when counted with the sign and if i is even, we count minus two flow lines. Therefore the counting of flow lines noted by $n(p,q)$ gives:

$$n(e_{i+1}, e_i) = (-1)^i - 1\,,$$

and the Morse cohomology boundary operator equals:

$$\partial_i([e_i]) = n([e_{i-1}], e_i)[e_{i+1}] = ((-1)^i - 1)[e_{i+1}]$$

□

Now for even n, EM_1 of the form:

$$0 \qquad \langle[e_0]\rangle \xrightarrow{0} \langle[e_1]\rangle \xrightarrow{-2} \langle[e_2]\rangle \xrightarrow{0} \cdots \xrightarrow{-2} \langle[e_n]\rangle \xrightarrow{0} 0$$

and for odd n:

$$0 \qquad \langle[e_0]\rangle \xrightarrow{0} \langle[e_1]\rangle \xrightarrow{-2} \langle[e_2]\rangle \xrightarrow{0} \cdots \xrightarrow{0} \langle[e_n]\rangle \xrightarrow{0} 0$$

Hence we have the EM_2 pages as follows:

$$0 \qquad \langle[e_0]\rangle \qquad 0 \qquad \frac{\langle[e_2]\rangle}{\langle-2[e_2]\rangle} \qquad 0 \qquad \cdots \qquad \frac{\langle[e_n]\rangle}{\langle-2[e_n]\rangle} \qquad 0$$

if n is even, and in the odd case:

$$0 \qquad \langle[e_0]\rangle \qquad 0 \qquad \frac{\langle[e_2]\rangle}{\langle-2[e_2]\rangle} \qquad 012 \qquad \cdots \qquad \langle[e_n]\rangle \qquad 0$$

When it comes to higher differentials, we notice the following: Only for odd r the differentials are possible nontrivial Hence for $\mathbb{RP}^2$ we are done and we can get the complex k-groups: For $K^0(\mathbb{RP}^2)$ we get a associeted graded group that is isomorphic to $\mathbb{Z} \oplus \mathbb{Z}_2$. Hence we need to calculate the extension

$$0 \longrightarrow \mathbb{Z} \longrightarrow K^0(\mathbb{RP}^2) \longrightarrow \mathbb{Z}_2 \longrightarrow 0$$

giving us $K^0(\mathbb{RP}^2) \cong \mathbb{Z} \oplus \mathbb{Z}_2$. And similar $K^1(\mathbb{RP}^2) = 0$. To understand the higher differentials we need to get specific about the used filtration. We define the filtration inductively as follows: For any i define $c_i \in (\lambda_i, \lambda_{i+1})$ and with this we define the filtration

$$N_{-1} := \varnothing, \quad N_i = \tilde{\Lambda}(-\infty, c_i) \text{ the sublevel sets }, \quad N_n := \mathbb{RP}^n\,.$$

With this filtration it is somewhat easy to see that (N_i, N_{i-1}) gives a regular index pair of $[e_i]$. One just needs to go through the definitions: First it is clear, that the only critical point inside of the closure of $N_i \smallsetminus N_{i-1}$ is $[e_i]$ and it is in the interiour. Secondly, the positiv lime invariance is clear, since $\tilde{\Lambda}$ decreases along flow lines. And N_{i-1} is an exit set also by definition. Finally the cofibration property of the pair N_i, N_{i-1} is to be shown. For this we notice, that c_i is a regular value of R. Hence there exists an $\varepsilon > 0$, such that $f^{-1}([a, a+\varepsilon))$ contains no critical point and hence this defines an open color of N_i inside N_{i+1}. This open color can be shown to be an NDR pair, which in turn is a cofibration. Those are standart arguments seen in [Whi78] Chapter 5.1.

Lemma 6.0.7. *In the filtration, each N_i is homotopically equivalent to $\mathbb{RP}^i$.*

Proof. Notice how for all i the sublevelset $\tilde{\Lambda}^{-1}(-\infty, \lambda_i]$ is an embedded projective space $\mathbb{RP}^i$.To show that N_i is homotopically equivalent we start by noticing that $N_i \simeq \tilde{\Lambda}^{-1}(-\infty, \lambda_{i-1} + \varepsilon)$ for all $\varepsilon > 0$ small enough. Now the next step is to get rid of epsilon. Let L_ε denote $\tilde{\Lambda}^{-1}(-\infty, \lambda_{i-1} + \varepsilon)$ and hence we want to show that $L\varepsilon \simeq L_0$. For this we define

$$H : L_\varepsilon \times [0,1] \to L_\varepsilon \quad ([x], t) \text{ with } \|x\| = 1 \mapsto \left[\sum_{j=1}^{i} x_j e_j + (1-t) \sum_{j>i} x_j e_j\right]$$

First we need to check if this map is well defined in two senses. Is the definition invariant under the chosen representative and do we map into L_ε. For the independence of the chosen representative we notice that the image of x and $-x$ differ by sign and hence define the same equivalence class. Also applying $\tilde{\Lambda}$ to the image we notice that the image is less then $\lambda_i + \varepsilon$. this is true, since $\tilde{\Lambda}$ is a weighted sum of the eigenvalues and the addition of $(1-t)$ decreases the weights on the bigger eigenvalues reducing the overall weighted sum.

in formeln nachrechnen

Now obviously H_1 maps to $\tilde{\Lambda}{-}1(-\infty, \lambda_i]$. ∎

Now that we have our filtration we can start looking at higher differentials. If n is even, then every other group in EM_2 is zero hence all higher differentials vanish as only odd d_r are candidates to be non trivial. So let n be odd. Then only differentials landing in the last group EM_n can possible be non trivial. However, for all $r < n$ the differential that lands in $EM_r^m \cong \mathbb{Z}$ starts in $EM_r^{m-r} \cong \mathbb{Z}_2$. Hence they are all trivial as well. Now if $r = m$ we have the differential

d_3^0 ist sowieso 0, da das bild 0 ist!!

$$\mathrm{d}_m^0 : EM_m^0 \to EM_m^m$$

which is possible non-zero. However this differential is given as follows:

$$\begin{tikzcd}
 & & K^0(N_m) \arrow[d, "i^*"] \\
K^0(N_0, \varnothing) \arrow[r, "\gamma"] \arrow[drr, "d_m^0"] & K^0(N_0) \arrow[r, "\widetilde{\alpha^{-m}}"] & K^0(N_{m-1}) \arrow[d, "\delta"] \\
 & & K^1(N_m, N_{m-1})
\end{tikzcd}$$

Here α^{-m} is any left inverse of α^m or taking the preimage (the well definednes or rather independence of the choosen inverse was shown in the definition of the derived exact couple). The vertical sequence is exact and since $N_{m-1} = N_m \setminus D$ where D is contractible, we have that i^* is surjective and hence δ is trivial and thereby $dM_m^0 = 0$. Then For this we start with $n = 3$ and then go up. In the case of $n = 3$ we are trivial, since d_3^0 is given by the composition

$$\begin{tikzcd}
& & K^0(N_3) \arrow[d, "i^*"] \\
K^0(N_0,\varnothing) \arrow[r, "\gamma"] \arrow[drr, "d_3"] & K^0(N_0) \arrow[r, "\widetilde{\alpha^{-2}}"] & K^0(N_2) \arrow[d, "\delta"] \\
& & K^1(N_3,N_2)
\end{tikzcd}$$

Here α^{-1} is any left inverse of α^2 or taking the preimage (the well definednes or rather independence of the choosen inverse was shown in the definition of the derived exact couple). The vertical sequence is exact and since $N_2 = N_3 \setminus D$ where D is contractible. A contraction is given by flowing along the positive gradient of $\tilde{\Lambda}$. Hence, we have that i^* is surjective and thereby δ is trivial resulting in $dM_m^0 = 0$. For all real projective spaces the spectral sequence becomes stable on the second page and hence we can calculate the complex K-groups of $\mathbb{RP}^n$: For even n we have the two graded associated groups:

$$\begin{aligned}
\mathrm{gr}(K^0(\mathbb{RP}^n)) &= \mathbb{Z} \oplus \underbrace{\mathbb{Z}_2 \oplus \cdots \oplus \mathbb{Z}_2}_{\frac{n}{2}\text{summands}} \\
\mathrm{gr}(K^1(\mathbb{RP}^n)) &= 0
\end{aligned}$$

Example 6.0.8. Make a produkt morse funktion on $\mathbb{RP}^2 \times \mathbb{RP}^4$!.

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